



RuleCOSI+: Rule extraction for interpreting classification tree ensembles

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ABSTRACT

Despite the advent of novel neural network architectures, tree-based ensemble algorithms such as random forests and gradient boosting machines still prevail in many practical machine learning problems in manufacturing, financial, and medical domains. However, tree ensembles have the limitation that the internal decision mechanisms of complex models are difficult to understand. Therefore, we present a post-hoc interpretation approach for classification tree ensembles. The proposed method, RuleCOSI+, extracts simple rules from tree ensembles by greedily combining and simplifying their base trees. Compared with its previous version, RuleCOSI, this new version can be applied to both bagging (e.g., random forest, RF) and boosting ensembles (e.g., gradient boosting machines, GBM) and run much faster for ensembles with hundreds of trees. To assess the performance and applicability of the method, empirical experiments were conducted using two bagging algorithms and four gradient boosting algorithms over 33 datasets. RuleCOSI+ could generate the best classification rulesets in terms of F-measure together with RuleFit for RF and GBM models of the datasets among five ensemble simplification algorithms, but the rulesets of RuleCOSI+ had, on average, less than half the size of those of RuleFit. Moreover, RuleCOSI+ had the best antecedent uniqueness rate (“UNIQ”) among the five algorithms, and had also ranked high in the number of rules (“NRULES”) and the rule reduction rate (“REDU”). In addition, the proposed method could reduce generalization errors in the simplified rulesets to obtain, on average, slightly better classification errors than original models of two bagging and three gradient boosting algorithms except CATBoost.

1. Introduction

The idea of ensemble learning is to aggregate the predictions of multiple models to improve the performance of every individual model forming the ensemble. This idea has been present for over 30 years. Since its introduction, several theoretically sound techniques for applying it have been presented. These techniques can be divided into two main construction frameworks: independent ensembles, such as bagging and random forests (RFs), and dependent ensembles, such as gradient boosting machines (GBM) [1]. When the base learners in the ensemble are trees, they are called tree ensembles or decision forests. Tree ensembles are widely used not only in academia and research, but also in practical applications in diverse domains, such as in manufacturing, finance, healthcare, and biology. The results obtained using ensembles have a good record of being accurate and showing low bias and variance [2]. However, one drawback of using tree ensembles is that these models generally behave as black box models. This is because tree ensembles tend to obtain better results as the number of trees in the

ensemble increases, sometimes to hundreds of trees. Moreover, the complexity of the models is further increased if one considers the different prediction aggregation mechanisms used for the various types of ensembles.

The issue of model transparency has become prevalent in recent years in the field of machine learning, because for certain specific practical applications (such as medical diagnosis, credit card fraud detection, loan approval applications, or fault detection in manufacturing processes), there is a need to understand the decision mechanisms of the predictive models, as well as to provide explanations for the decisions of the models for specific examples in the data. Indeed, one part of the field of explainable artificial intelligence studies the inherent interpretability of a model, or the creation of methods able to take black box models and generate simpler models for providing explanations of the decisions of the source model [3]. Under these definitions, we can see that tree ensembles are not inherently interpretable, because the ensembles tend to contain dozens, hundreds, or even thousands of trees. Therefore, one common approach is to derive an

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interpretable model from a tree ensemble. These techniques are known as ensemble post-hoc explainability simplification methods. These simplified explanation models should be transparent models, such as a single decision tree (DT), decision rule set, or linear model.

In recent years, several studies have been conducted to address this problem. The *RuleFit* (RFIT) algorithm [4] generates tree ensembles that are transformed into rules and fitted with a LASSO linear regressor to find the weights of the rules and their interactions. The *inTrees* (IT) framework [5] extracts, measures, prunes, and selects rules from tree ensembles and ultimately summarizes them into a simplified tree ensemble learner (STEL) that can be used to make predictions. Another approach called *DefragTrees* (DFT) [6] uses a Bayesian model selection method and a small set of rules to obtain a simpler representation of the ensemble that is essentially equivalent to the original ensemble model. However, these approaches have some drawbacks. For example, the RFIT algorithm generates very accurate rulesets, but they tend to have many rules and are therefore not interpretable. For their part, the IT framework and DFT algorithm generate rulesets that have redundant information in the form of repeated conditions, which could potentially reduce its interpretability. *Forest-based trees* (FBTs) [7,8] are another recently proposed approach that aims to make tree ensembles interpretable by generating a single DT that approximates the predictions of the original ensemble. The trees generated by FBTs are certainly more interpretable than a tree ensemble, but the size of the model tends to be larger than that of a typical DT. When analyzing simplified models generated by the previously introduced methods, a notorious tradeoff remains between the classification performance and interpretability. Accordingly, in our previous study, we presented RuleCOSI, a combination and simplification algorithm for reducing this tradeoff for imbalanced classification [9]. However, the method was rather limited in its application because it only worked with imbalanced data and AdaBoost-based small trees.

Therefore, in this paper, we present RuleCOSI+, which stands for "Rule Combination and Simplification," a novel and fast post-hoc explainability method for generating a rule-based classifier from a tree ensemble. The proposed method uses the splits provided by the tree ensemble without considering the hierarchy inside each tree, and in most cases, can greedily construct a decision list that can approximate the classification performance of the original tree ensemble. This method works with both independent and dependent tree ensembles. To assess the applicability of our proposed method, we conducted several experiments using six tree ensemble algorithms and 33 well-known datasets from the University of California Irvine (UCI) machine learning repository. Additionally, we introduce two simple measures to aid in comparing ensemble post-hoc explainability methods. The first measure is called *REDU*, and computes the amount of the size reduction obtained by the simplification method. The second measure is called *UNIQ*, and quantifies the amount of non-redundant information contained in the model. Using this framework, we compare our method with four state-of-the-art ensemble post-hoc simplification methods and three benchmark interpretable models. The results show that RuleCOSI+ is able to maintain the ensemble performance in most cases, and generally outperforms other similar methods in terms of *REDU* and *UNIQ*.

The contributions of the study are as follows.

- The development of a novel tree ensemble post-hoc explainability method that significantly reduces the size of large ensembles, produces interpretable models with non-redundant information, and maintains a good classification performance in most cases. RuleCOSI+ showed the best F-measure values for the RF and GBM models of 33 datasets among five ensemble simplification algorithms, with small and non-redundant ruleset.
- The broad applicability of the method, that is, its application to both dependent and independent types of tree ensembles. RuleCOSI+ was implemented by extending the previous version, RuleCOSI, to be able

to work for both bagging (e.g., RF) and boosting ensembles (e.g., GBM)

- The introduction of a framework for evaluating the performance of ensemble post-hoc simplification methods, in terms of the size reduction (*REDU*) and non-redundant interpretability (*UNIQ*) of the generated models. Five ensemble simplification algorithms were compared with the aforementioned measures. RuleCOSI+ could get the highest *UNIQ* and the second highest *REDU* values, but it had the best F-measure values, on average, over 33 datasets.

The background concepts regarding ensembles, explainability in machine learning, rule-based classifiers, and related works are introduced in Section 2. Section 3 describes the proposed method in detail. Section 4 introduces the two new interpretability measures. Section 5 describes the experimental design, and Section 6 presents the experimental results. Section 7 provides an interesting discussion of the results. Finally, Section 8 concludes the paper, stating its limitations and future work.

2. Background

Before presenting the proposed method, we introduce the basic notations required to understand the algorithm. Then, we briefly discuss the issue of interpretability for ensemble models and discuss related ensemble post-hoc explainability methods.

2.1. Classification tree ensembles

A classification tree ensemble H is a set of classification trees $h_k \in H$, with $k = 1$ to K . We will simply refer to them as "tree ensembles" and "trees," respectively. Each tree h_k is constructed from a training set with n samples and m features, i.e., $D = \{(x_i, y_i)\}$ with $i = 1$ to n , $x \in \mathbb{R}^m$, and $y \in \mathbb{R}$. We restrict our analysis to the binary classification task, i.e., $y \in \{0, 1\}$ —but it can be trivially extended to a multiclass classification task. The trees in H can be constructed independently, as in bagging trees or RFs, or dependently, as in Gradient Boosting Trees (GBTs) [11].

When the tree ensemble H is used to classify new instances, it takes the output of each tree function $h_k \in H$ and uses an aggregation function ϕ , such that $H(x) = \phi(h_1, h_2, \dots, h_K)$. The aggregation function differs depending on the ensemble construction framework.

For independent tree ensembles, in which the output of a tree h_k is a vector containing the class probabilities, and ϕ_{ind} is the average over the class probability vector obtained from each tree function h_k , i.e., $\phi_{ind}(h_1, \dots, h_K) = \frac{1}{K} \sum_{k=1}^K h_k$.

For dependent tree ensembles, in which the output of a tree h_k is the log-odds of the probability of predicting one of the two classes, and ϕ_{ind} is the total sum of each tree function output h_k , i.e., $\phi_{dep}(h_1, \dots, h_K) = \sum_{k=1}^K h_k$. The output of ϕ_{dep} is posteriorly transformed into a class probability using a logistic function.

2.2. Decision rules

A decision rule is an IF (*Conditions*)–THEN (y) expression, where y is the class label and *Conditions* are a conjunction of logical conditions [10]. Therefore, we formally define that a rule has the form $r: A \rightarrow y$, where $A = \text{Conditions}$, which is a logical conjunction of the conditions $a \in A$, called the rule antecedent or body of the rule, and y is the rule consequent class label, also called the head of the rule. The set of conditions A of a rule r , denoted by $r.A$, encloses a region π_r in the feature space $\mathbb{R}^m$ of the training set D . Rule r covers a record x if $x \in \pi_r$.

The rules are grouped into a ruleset R . When classifying a sample, R works in one of two ways: either all of the rules $r \in R$ are evaluated, the result of every rule is aggregated with a voting or averaging mechanism; or the rules are evaluated one-by-one until one rule is fired; in this latter case, the head of the fired rule is used as the model prediction. The latter

is also called the decision list and is used as the output of the method presented herein.

The *coverage* of a rule r over D is given by the fraction of records in D that fall in the feature space region defined by the body of the rule, i.e., $\text{cov}(r) = |r.A|/n$, where $|r.A|$ is the number of records satisfying the body of rule r , and $n = |D|$. The *accuracy* or *confidence* of a rule r over D is the fraction of the records covered by r that also agree with the head of the rule, i.e., $\text{conf}(r) = |r.A \cap r.y|/|r.A|$, where $|r.A \cap r.y|$ is the number of records satisfying both the body and the head of the rule r .

2.3. Interpretation of tree ensembles

The interpretability of a model refers on how much sense the model makes to a human, whereas the explainability of a model is associated with an explainable interface between the humans and model [3]. Some types of explainable interfaces are used to uncover the internal mechanisms of a tree ensemble. Traditionally, this is mainly done using feature importance (FI) methods, which rank the features or variables used in the model according to their importance when constructing each of the trees in the ensemble [11]. FI techniques can shed some light on which features have more influence on the classification decision of the model, but the ranking is rather limited, as it is unable to provide more specific information (such as feature interactions), or the classification boundaries on the feature space for making different decisions. One solution to this problem is to generate separate explainable models, usually transparent models, such as DTs or decision rules. These explainable models are categorized as ensemble post-hoc explainability models. RuleCOSI+ belongs to this category.

Post-hoc explainability methods for tree ensembles typically generate rule-based classifiers or single DTs. The RFIT algorithm [4] constructs a tree ensemble that is posteriorly transformed into a set of rules containing one rule for each leaf node on the original trees. Subsequently, each rule is converted into a new feature using dummy encoding, and a LASSO regressor is fitted to filter the most important features and rules. Although this approach produces a transparent model with the addition of a weight to denote the rule importance, the rulesets tend to be large, so the results remain difficult to interpret. REEMTIC [12] is a rule extraction algorithm that uses an ensemble of DTs as a starting point, after which several decision rules are extracted and combined through logic minimization techniques. The result is a ruleset in the disjunctive normal form—a disjunction of conjunctions. The proposed method uses a combination heuristic and a pessimistic error for rule pruning, which is a more practical method. A rather different approach is the "GENetic Extraction of a Single Interpretable Model (GENESIM)" method [13]. This method uses a genetic approach to examine different combinations of subsets from the original tree ensemble, with the purpose of not only obtaining an interpretable model, but also of enhancing its predictive performance. The results show that the simplified model outperforms most of the other tree learning algorithms; however, it cannot be compared with the ensemble results. Additionally, certain results in the experiments showed a high average number of nodes per model (e.g., 103.15 nodes for the *cars* dataset), and thus cannot be considered as interpretable. Finally, the execution time is often expected to be high, owing to the genetic nature of the method. The DFT method [6] adopts a probabilistic representation of a tree ensemble and applies a Bayesian model selection approach to optimize the number of regions, their shapes, and locations by estimating the parameters of the probabilistic model. The result is a simplified model "equivalent" to the original ensemble. In general, the number of rules should be carefully chosen to balance the tradeoff between the interpretability and accuracy of these models. The IT [5] framework extracts rules from tree ensembles, and then prunes the rules based on a balance between the complexity and error rate of each rule. Finally, redundant rules are pruned using feature selection techniques; thus, the rules are summarized into a STEL. The results from DFT and IT can be compared with those from other tree learners, but not against the

original ensemble.

Tree ensembles can also be summarized as a set of fuzzy IF-THEN rules as studied in a few works. The method presented in [14] generates linguistic based fuzzy rules using different group of attributes and different projections from the data. The prediction is done using fuzzy reasoning and the interpretability is enhanced by grouping antecedents that are highly correlated in the same rule, thus reducing the size. More recently, in [15] a rule inference engine is used to combine the classifiers generated by heterogeneous ensembles, capturing the uncertainty in the outputs of the base classifiers. After that, fuzzy rules are generated on the outputs each base classifier, seeing the output as the class posterior probability of the observations.

Other type of methods that are related with RuleCOSI+ are the surrogate model approximation methods. These methods find a simpler surrogate model for the ensemble, and simultaneously, the surrogate model should approximate the performance relative to that of the original ensemble. *Born-again tree ensembles* [16] search for a DT T of a minimal size that is faithful to the tree ensemble. In other words, the tree function $T(x_i) = H(x_i)$ for all $x_i \in \mathbb{R}^m$. As expected, finding that identifying a minimal tree is an NP-hard problem, the method uses a dynamic-programming-based algorithm to generate pruning and bounding rules to reduce the execution time. This method has only been tested using RFs. FBTs [7,8] is a method that starts by extracting a set of rule conjunctions from a tree ensemble. Then, the rule conjunctions are combined, and redundant conjunctions are eliminated. For each conjunction, the class probabilities are computed and used to prune conjunctions with a low probability. Finally, all the best rule conjunctions are organized in a tree structure by finding the best splits at each level, using a procedure based on information gain and entropy.

3. RuleCOSI+: Enhanced rule combination and simplification

The core ideas of our study are presented in this section. We begin by formally defining the research problem being solved in this work and present an overview of the main idea behind the proposed method. Next, the steps of the algorithm are described with sufficient detail, so that the readers can understand the combination and simplification procedures. We conclude the section by providing some details of the specific algorithm implementation adopted while conducting the experiments.

3.1. Problem definition

The problem addressed in this study is the generation of interpretable models from tree ensembles. For this, we propose a post-hoc explainability method for generating a rule-based classifier to approximate the predictions of the original ensemble. The problem is defined as follows.

Post-hoc rule extraction from tree ensembles. A rule extraction function $G(H)$ takes a trained tree ensemble model $H(x)$ and generates the ordered set of rules, $R = G(H)$, such that the following holds.

$$\forall x \in \mathbb{R}^m, R(x) \approx H(x) \quad (1)$$

The equation implies the ruleset R generated by $G(H)$ produces the same prediction as H for x in training set D . The function $G(H)$ is required to consider the aggregation function ϕ associated with H when extracting the rules. The output of the function should be an ordered set of rules, also known as decision lists, because the interpretation is simpler than that of an unordered ruleset.

3.2. Algorithm overview

RuleCOSI+ is a rule extraction algorithm that greedily searches among the set of all possible combinations of the feature splits obtained from the tree ensemble H . The algorithm starts by extracting a raw ruleset from each tree in H . After the simplified ruleset R° is initialized with the first ruleset R_1 , a greedy optimization search is performed. In

Table 1

Summary of 33 datasets used for the experiments.

Dataset	Class (minor; major)	Imbalanced ratio (IR)	# inst.	# atts.
australian	(approved; not approved)	1.24	690	14
bands	(band; noband)	1.7	365	19
banknote	(genuine; forged)	1.25	1,372	4
biodeg	(ready; not ready)	1.96	1,055	40
cardiotcg	(presence; absence)	3.51	2,126	21
credit	(default; not default)	11.11	666	51
cryotherapy	(yes; no)	1.14	90	6
divorce	(yes; no)	1.02	170	54
ecoli0vs1	(im; cp)	1.86	220	7
ecoli1	(im; remainder)	3.36	336	7
ecoli2	(pp; remainder)	5.46	336	7
ecoli3	(imU; remainder)	8.6	336	7
glass	(non-window glass; remainder)	3.2	214	9
glass0	(build-win-float-proc; remainder)	2.06	214	9
glass1	(build-win-non_float-proc; remainder)	1.82	214	9
glass6	(headlamps; remainder)	6.38	214	9
haberman	(5yr.surv.; not surv.)	2.77	303	3
heart	(absence; presence)	1.25	270	13
ionosphere	(bad; good)	1.78	351	34
maintenance	(failure; normal)	28.50	10,000	6
mammographic	(benign; malignant)	1.06	830	5
new-thyroid1	(hyper; remainder)	5.14	215	5
new-thyroid2	(hypo; remainder)	4.92	215	5
occupancy	(not; occupied)	3.71	8,143	5
pageblocks	(valid; cum)	8.79	5,472	10
pima	(positive; negative)	1.9	768	8
segment0	(brickface; remainder)	6.02	2,308	19
tic-tac-toe	(positive; negative)	1.29	958	27
vehicle2	(bus; remainder)	2.52	846	18
vehicle3	(opel; remainder)	2.52	846	18
wisconsin	(malignant; benign)	1.86	683	9
yeast1	(nuc; remainder)	2.46	1,484	8
yeast3	(me3; remainder)	8.11	1,484	8

Table 2

Hyperparameter values explored with the grid search for simplification and baseline algorithms used in the experiments.

Category	Algorithm	Parameters
Ensemble simplification algorithms	RuleCOSI+ (RC+)	$\alpha \in \{0.5, 0.95\}, \beta \in \{0, 0.01, 0.1\}, c \in \{10\%, 25\%, 40\%\}$
	RuleFit (RFIT)	$\alpha \in \{0.1, 1, 10, 100\}, \text{include linear} \in \{\text{True}, \text{False}\}$
	inTrees (IT)	$\text{maxdecay} \in \{0.05, 0.01\}, \text{minfreq} \in \{0, 0.1, 0.01\}$
	DefragTrees (DFT)	$\text{maxitr} \in \{50, 100\}, \text{maxrules} \in \{4, 8, 16\}$
Baseline rule generation algorithms	Forest-based Tree (FBT)	$d \in \{2, 3, 4, 6\}, \text{pruning} \in \{\text{no}, \text{AUC}\}, \text{conjunctions} = 250$
	Decision Tree (DT)	$d \in \{\text{no limit}, 2, 3, 4, 6\}, \alpha \in \{0, 0.005, 0.01\}$
	SkopeRules (SKR)	$\text{min_precision} \in \{0.5, 0.95\}, n_trees \in \{4, 8, 20, 50\}, d \in \{2, 3, 4, 5, 6\}$
	BoostedRules (BSTR)	$\text{maxrules} \in \{4, 8, 16, 32\}$

each iteration, it creates new rules by merging existing rules with the COMBINERULESETS function, and then prunes and generalizes rulesets to avoid overfitting using two procedures, SEQUENTIALCOVERINGPRUNING and GENERALIZERULESET. At the end of each iteration, better ruleset between the new combined ruleset and the previous ruleset is maintained. The final step is to append a default rule to the simplified ruleset R° .

Algorithm. RuleCOSI+

Input: tree ensemble (H), training set (D), confidence threshold (α), coverage threshold (β), and confidence level (c)

(continued on next column)

(continued)

Output: simplified ruleset R°	
1:	(Extract raw rulesets from the tree ensemble.) $R^* \leftarrow R_k R_k = \psi(h_k) \forall h_k \in H$
2:	(Initialize a simplified ruleset.) $R^\circ \leftarrow \text{SimplifyRuleset}(R_1^*, \alpha, \beta, D)$
3:	(Start greedy search loop.) for $k = 2$ to K do
4:	(Combine rulesets.) $R' \leftarrow \text{CombineRulesets}(R^\circ, R_k, \alpha, D)$
5:	(Prune rulesets.) $R' \leftarrow \text{SequentialCoveringPruning}(R', \alpha, D)$
6:	(Generalize rulesets.) $R' \leftarrow \text{GeneralizeRuleset}(R', \alpha, \beta, c, D)$
7:	(Compare model performances.) Choose the better model between R° and R'
8:	(Early stopping criterion.) if R° has perfect performance then exit the loop
9:	Add a default rule to R° predicting the majority class among uncovered samples
10:	(Return the simplified ruleset.) return R°

The proposed method is based on our previous work [9]. RuleCOSI+ takes the same basic structure—extract, combine, and simplify rules—but adopts a different combination approach. As a result, the new version can handle tree ensembles with hundreds of trees, as well as work with both independent and dependent ensemble types. Additionally, the simplified rules generated by the new algorithm are considered soft rules, as they provide class probability vectors in addition to hard predictions.

The main functions are explained and discussed with more detail in the following subsections.

3.3. Ruleset extraction from dependent and independent tree ensembles

The rule extraction process from a tree h_k is straightforward. We consider that there is a function ψ that can transform a tree h into a ruleset R , i.e., $R = \psi(h)$, such that the conjunction of all the decision nodes from the root node until each terminal node $t \in h$ is transformed into a decision rule (see line 1 of the RuleCOSI+ algorithm). However, depending on the ensemble construction framework, i.e., whether it is independent or dependent, the extraction process can be slightly different. For this reason, the rules in R store additional information extracted from their source tree in the form of *rule meta-attributes* that are accessed using the dot notation. For example, $r.attr_1$ refers to the meta-attribute $attr_1$ of rule r .

Independent tree ensembles such as GBT comprise tree functions that output a logit scores, because the regression trees $h_k \in H$ perform a logistic regression. The class probability vector of the rule can be constructed using the logistic function. Specifically, for a rule r , we first obtain the probability $p_{r,c}$ of class $c = 1$ using the logit score of r , $r.logit$; this is $p_{r,1} = 1/(1 + \exp(-r.logit))$. Then, we construct the class probability vector, $r.p \leftarrow (1 - p_{r,1}, p_{r,1})$.

Dependent tree ensembles, such as bagging and RFs, provide the sample distribution in the leaf nodes; this can be transformed into a class probability vector by normalizing the values. Therefore, when a rule r is extracted from a dependent tree, the normalized sample distribution is stored in the p meta-attribute of the rule.

3.4. Ruleset combination

The goal of this stage is to grow new rules through the exploration of the diverse combinations of covered regions from the feature spaces of the rules in rulesets R_1 and R_2 , with the restriction imposed by the parameter α (the rule confidence threshold).

The COMBINERULESETS function executes the ruleset combination procedure. All of the ordered pairs (r_1, r_2) obtained from the Cartesian product $R_1 \times R_2$ are individually combined (see line 2). The individual combination of two rules, r_1 and r_2 , generates a new combined rule r' . Consequently, the covered regions of r_1 and r_2 intersect, i.e., $\pi_{r'} = \pi_{r_1} \cap \pi_{r_2}$. The first intuitive restriction is that $\pi_{r'} \neq \emptyset$, that is, the covered region of the new rule should not be null. The second restriction is that the confidence of the new rule should be greater than α . This ensures that only accurate rules are considered in the remaining steps of the

Table 3

Comparison of F-measures ($\mu \pm \sigma$) among ensemble simplification algorithms for RF in 3×10 fold CV. A symbol marking indicates the result of the Wilcoxon signed-rank test, i.e., $\uparrow$: better, $\downarrow$: worse, and no mark: not statistically different (p-value $> .05$).

Dataset	RF	RF-RFIT	RF-IT	RF-DFT	RF-FBT	RF-RC$\pm$	SKR	DT
australian	84.71 $\pm$ 4.56	82.91 $\pm$ 5.69 $\downarrow$	73.34 $\pm$ 25.48 $\downarrow$	82.96 $\pm$ 3.83 $\downarrow$	83.08 $\pm$ 4.87	83.33 $\pm$ 4.34	56.20 $\pm$ 26.19 $\downarrow$	84.66 $\pm$ 2.99 $\downarrow$
bands	80.22 $\pm$ 4.01	77.16 $\pm$ 2.18 $\downarrow$	73.66 $\pm$ 8.46 $\downarrow$	74.52 $\pm$ 6.02 $\downarrow$	76.40 $\pm$ 2.57 $\downarrow$	76.47 $\pm$ 5.94 $\downarrow$	9.49 $\pm$ 14.35 $\downarrow$	70.49 $\pm$ 5.95 $\downarrow$
banknote	99.05 $\pm$ 0.64	99.48 $\pm$ 0.66 $\uparrow$	98.58 $\pm$ 1.16	95.76 $\pm$ 1.75 $\downarrow$	97.47 $\pm$ 3.53 $\downarrow$	98.72 $\pm$ 0.93	70.98 $\pm$ 23.63 $\downarrow$	97.85 $\pm$ 1.50 $\downarrow$
biodeg	89.81 $\pm$ 2.79	88.37 $\pm$ 2.46 $\downarrow$	88.06 $\pm$ 2.68 $\downarrow$	83.59 $\pm$ 3.37 $\downarrow$	83.91 $\pm$ 4.20 $\downarrow$	88.40 $\pm$ 3.17 $\downarrow$	16.67 $\pm$ 27.00 $\downarrow$	86.47 $\pm$ 2.77 $\downarrow$
cardiotcg	96.10 $\pm$ 0.90	96.17 $\pm$ 1.01	96.25 $\pm$ 1.07	94.04 $\pm$ 1.08 $\downarrow$	90.97 $\pm$ 2.28 $\downarrow$	96.36 $\pm$ 1.14	86.88 $\pm$ 16.51 $\downarrow$	95.78 $\pm$ 1.17 $\downarrow$
credit	94.72 $\pm$ 6.31	98.47 $\pm$ 4.37 $\uparrow$	97.23 $\pm$ 9.40 $\uparrow$	97.47 $\pm$ 4.33	99.14 $\pm$ 2.64 $\uparrow$	98.83 $\pm$ 3.03 $\uparrow$	96.17 $\pm$ 12.42	99.14 $\pm$ 2.64 $\uparrow$
cryotherapy	93.19 $\pm$ 9.25	91.19 $\pm$ 10.59	86.92 $\pm$ 12.43 $\downarrow$	86.91 $\pm$ 10.41 $\downarrow$	88.30 $\pm$ 7.64 $\downarrow$	87.71 $\pm$ 11.88 $\downarrow$	70.50 $\pm$ 28.56 $\downarrow$	89.97 $\pm$ 11.14
divorce	97.44 $\pm$ 3.20	97.24 $\pm$ 3.22	96.10 $\pm$ 4.80	97.24 $\pm$ 3.61	97.68 $\pm$ 3.11	96.66 $\pm$ 4.27	96.96 $\pm$ 4.15	97.90 $\pm$ 3.44
ecoli0vs1	99.22 $\pm$ 1.44	98.73 $\pm$ 1.95	98.28 $\pm$ 2.89	99.10 $\pm$ 1.51	98.87 $\pm$ 1.62	98.86 $\pm$ 1.65	97.02 $\pm$ 5.78 $\downarrow$	98.99 $\pm$ 1.57
ecoli1	93.08 $\pm$ 3.06	92.78 $\pm$ 3.67	92.85 $\pm$ 2.93	93.36 $\pm$ 3.11	93.59 $\pm$ 2.98	92.59 $\pm$ 3.55	83.88 $\pm$ 22.24 $\downarrow$	93.71 $\pm$ 3.50 $\downarrow$
ecoli2	96.98 $\pm$ 2.16	96.80 $\pm$ 1.85	96.05 $\pm$ 2.05 $\downarrow$	96.22 $\pm$ 1.96 $\downarrow$	96.52 $\pm$ 2.19	96.37 $\pm$ 1.82 $\downarrow$	17.90 $\pm$ 29.40 $\downarrow$	96.25 $\pm$ 2.13 $\downarrow$
ecoli3	96.10 $\pm$ 1.74	96.20 $\pm$ 1.62	95.39 $\pm$ 2.76	95.83 $\pm$ 1.84	94.99 $\pm$ 1.40 $\downarrow$	96.17 $\pm$ 1.75	69.00 $\pm$ 32.32 $\downarrow$	95.39 $\pm$ 1.74 $\downarrow$
glass	95.83 $\pm$ 3.25	95.73 $\pm$ 3.40	95.63 $\pm$ 3.87	95.12 $\pm$ 4.51	95.54 $\pm$ 3.71	95.87 $\pm$ 3.60	94.80 $\pm$ 3.96	95.99 $\pm$ 3.21
glass0	90.02 $\pm$ 7.72	87.35 $\pm$ 7.41 $\downarrow$	87.21 $\pm$ 6.04	83.10 $\pm$ 9.51 $\downarrow$	85.62 $\pm$ 6.40 $\downarrow$	89.70 $\pm$ 4.71	17.15 $\pm$ 22.77 $\downarrow$	85.49 $\pm$ 7.76 $\downarrow$
glass1	87.80 $\pm$ 5.28	82.35 $\pm$ 9.61 $\downarrow$	84.85 $\pm$ 6.54	79.04 $\pm$ 7.71 $\downarrow$	82.44 $\pm$ 6.44 $\downarrow$	85.87 $\pm$ 8.02	55.42 $\pm$ 30.56 $\downarrow$	80.21 $\pm$ 9.16 $\downarrow$
glass6	98.15 $\pm$ 1.85	97.95 $\pm$ 2.61	97.90 $\pm$ 2.07	97.42 $\pm$ 2.15 $\downarrow$	97.38 $\pm$ 2.45	97.53 $\pm$ 2.17	96.35 $\pm$ 3.44 $\downarrow$	96.33 $\pm$ 3.07 $\downarrow$
haberman	83.90 $\pm$ 2.60	84.81 $\pm$ 1.36	83.86 $\pm$ 5.66	84.66 $\pm$ 3.77	83.81 $\pm$ 3.08	84.54 $\pm$ 1.67	6.72 $\pm$ 10.96 $\downarrow$	83.81 $\pm$ 4.96
heart	87.24 $\pm$ 5.80	80.77 $\pm$ 8.49 $\downarrow$	77.16 $\pm$ 9.88 $\downarrow$	81.55 $\pm$ 7.62 $\downarrow$	80.11 $\pm$ 6.37 $\downarrow$	83.31 $\pm$ 6.18 $\downarrow$	25.05 $\pm$ 31.15 $\downarrow$	81.09 $\pm$ 8.60 $\downarrow$
ionosphere	95.03 $\pm$ 2.97	93.36 $\pm$ 3.66 $\downarrow$	92.43 $\pm$ 4.51 $\downarrow$	92.85 $\pm$ 4.14 $\downarrow$	89.90 $\pm$ 3.79 $\downarrow$	93.56 $\pm$ 3.29 $\downarrow$	89.47 $\pm$ 15.63 $\downarrow$	91.54 $\pm$ 3.31 $\downarrow$
maintenance	41.89 $\pm$ 8.51	61.85 $\pm$ 7.94 $\uparrow$	59.59 $\pm$ 10.92 $\uparrow$	35.94 $\pm$ 10.77 $\downarrow$	29.81 $\pm$ 15.26 $\downarrow$	62.99 $\pm$ 9.60 $\uparrow$	5.83 $\pm$ 7.42 $\downarrow$	69.58 $\pm$ 5.93 $\uparrow$
mammographic	83.30 $\pm$ 4.38	82.66 $\pm$ 5.19	22.56 $\pm$ 30.78 $\downarrow$	82.29 $\pm$ 4.65 $\downarrow$	83.47 $\pm$ 4.14	82.47 $\pm$ 4.31	45.03 $\pm$ 37.05 $\downarrow$	82.40 $\pm$ 4.83 $\downarrow$
newthyroid1	98.99 $\pm$ 1.56	98.90 $\pm$ 1.37	98.74 $\pm$ 1.87	98.62 $\pm$ 2.03	95.31 $\pm$ 3.22 $\downarrow$	98.33 $\pm$ 2.16	91.96 $\pm$ 17.94 $\downarrow$	97.66 $\pm$ 2.13 $\downarrow$
newthyroid2	99.02 $\pm$ 1.76	98.64 $\pm$ 2.00	98.07 $\pm$ 2.44	98.15 $\pm$ 1.84 $\downarrow$	97.67 $\pm$ 2.07 $\downarrow$	98.28 $\pm$ 2.17	93.58 $\pm$ 4.50 $\downarrow$	97.58 $\pm$ 2.28 $\downarrow$
occupancy	99.57 $\pm$ 0.14	99.58 $\pm$ 0.13	99.53 $\pm$ 0.15	99.21 $\pm$ 0.22 $\downarrow$	99.20 $\pm$ 0.23 $\downarrow$	99.58 $\pm$ 0.16	99.21 $\pm$ 0.21 $\downarrow$	99.57 $\pm$ 0.16
pageblocks	98.51 $\pm$ 0.36	98.36 $\pm$ 0.44 $\downarrow$	98.22 $\pm$ 0.39 $\downarrow$	97.60 $\pm$ 0.49 $\downarrow$	96.51 $\pm$ 1.26 $\downarrow$	98.50 $\pm$ 0.35	97.01 $\pm$ 0.89 $\downarrow$	98.38 $\pm$ 0.43 $\downarrow$
pima	82.98 $\pm$ 2.78	80.31 $\pm$ 3.45 $\downarrow$	78.45 $\pm$ 5.35 $\downarrow$	80.75 $\pm$ 3.34 $\downarrow$	81.72 $\pm$ 2.62 $\downarrow$	81.14 $\pm$ 3.00 $\downarrow$	34.69 $\pm$ 18.67 $\downarrow$	80.54 $\pm$ 4.09 $\downarrow$
segment0	99.80 $\pm$ 0.25	99.78 $\pm$ 0.24	99.75 $\pm$ 0.28	99.47 $\pm$ 0.34 $\downarrow$	93.60 $\pm$ 2.29 $\downarrow$	99.71 $\pm$ 0.24	83.96 $\pm$ 16.20 $\downarrow$	99.54 $\pm$ 0.35 $\downarrow$
tic-tac-toe	99.92 $\pm$ 0.46	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	99.47 $\pm$ 2.88	100.0 $\pm$ 0.00
vehicle2	98.89 $\pm$ 0.85	98.59 $\pm$ 1.05 $\downarrow$	97.95 $\pm$ 1.54 $\downarrow$	95.70 $\pm$ 2.20 $\downarrow$	90.16 $\pm$ 4.68 $\downarrow$	98.20 $\pm$ 1.73 $\downarrow$	26.28 $\pm$ 23.94 $\downarrow$	97.12 $\pm$ 1.59 $\downarrow$
vehicle3	87.69 $\pm$ 1.59	85.07 $\pm$ 1.58 $\downarrow$	86.45 $\pm$ 2.70 $\downarrow$	85.94 $\pm$ 2.32 $\downarrow$	85.87 $\pm$ 0.92 $\downarrow$	87.37 $\pm$ 1.83	14.56 $\pm$ 16.27 $\downarrow$	85.82 $\pm$ 1.75 $\downarrow$
wisconsin	97.80 $\pm$ 1.82	97.40 $\pm$ 1.83 $\downarrow$	96.00 $\pm$ 2.73 $\downarrow$	96.61 $\pm$ 2.30 $\downarrow$	94.33 $\pm$ 2.67 $\downarrow$	96.47 $\pm$ 2.01 $\downarrow$	95.65 $\pm$ 2.85 $\downarrow$	96.28 $\pm$ 2.03 $\downarrow$
yeast1	85.35 $\pm$ 1.65	82.84 $\pm$ 1.56 $\downarrow$	82.95 $\pm$ 2.58 $\downarrow$	83.28 $\pm$ 2.04 $\downarrow$	83.18 $\pm$ 2.02 $\downarrow$	84.38 $\pm$ 1.93 $\downarrow$	13.22 $\pm$ 13.42 $\downarrow$	83.13 $\pm$ 2.32 $\downarrow$
yeast3	97.15 $\pm$ 0.87	97.30 $\pm$ 0.87	95.91 $\pm$ 1.27 $\downarrow$	97.11 $\pm$ 0.86	97.11 $\pm$ 0.93	97.09 $\pm$ 0.98	40.76 $\pm$ 37.12 $\downarrow$	97.09 $\pm$ 1.13 $\downarrow$
Average	90.45 $\pm$ 2.90	91.49 $\pm$ 3.13 (improved)	88.66 $\pm$ 5.38	89.74 $\pm$ 3.50	89.20 $\pm$ 3.44	91.68 $\pm$ 3.14 (improved)	60.54 $\pm$ 16.98	91.08 $\pm$ 3.32

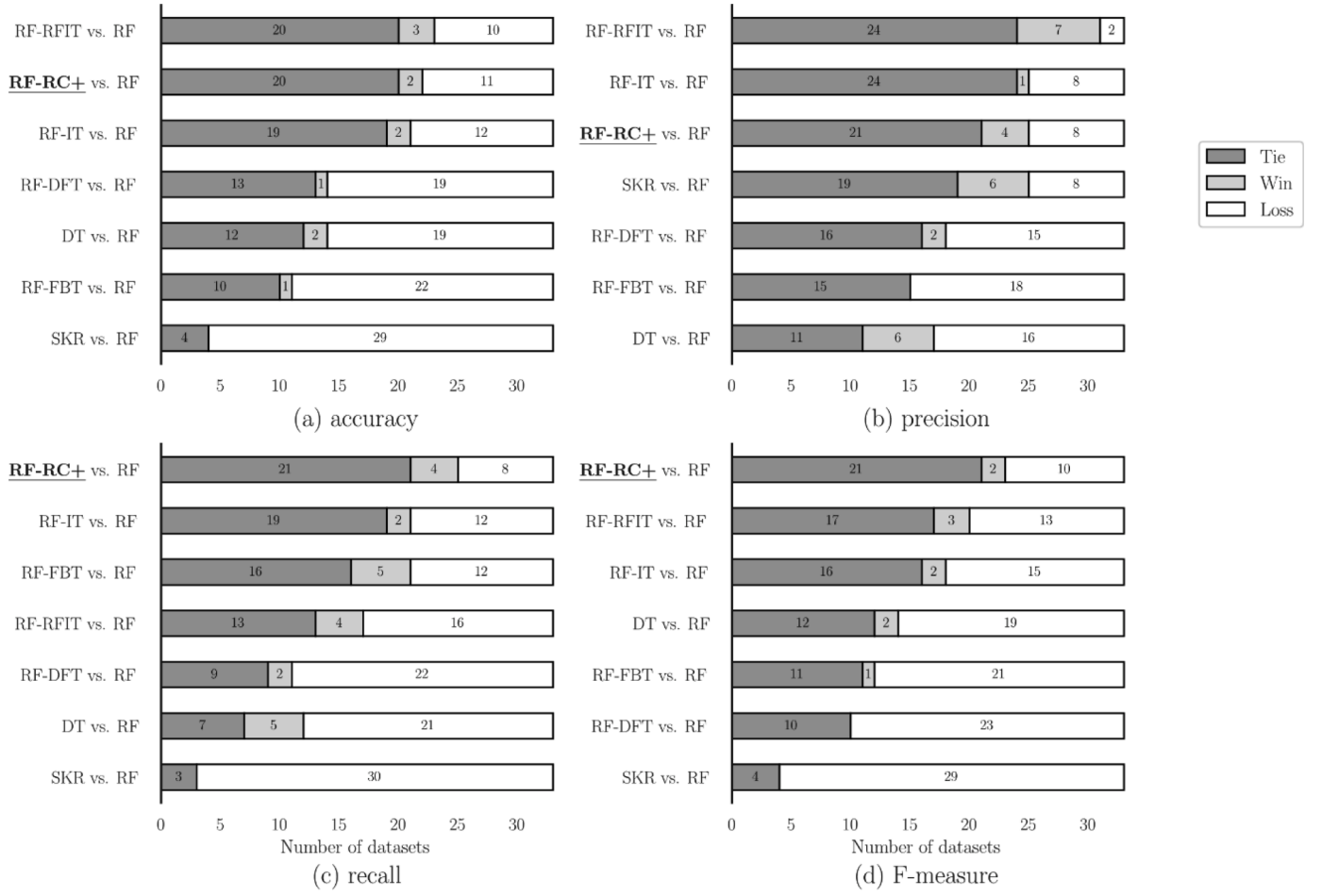


Fig. 1. Tie-win-loss record using the Wilcoxon signed-rank test among algorithms against original ensemble (RF) over 33 datasets.

process. The combination procedure is conceptually divided into two parts: the union of the rule conditions, and the class probability estimation.

In the first part, the condition sets of both rules are merged with the union operation (see line 4). Next, the new set of conditions is simplified to remove overlapping conditions (see line 5). We use the term “overlapping conditions” to refer to conditions in which the feature space of one condition is completely contained in the feature space of another condition. For example, if we have $a_1 = attr_1 > 5$ and $a_2 = attr_1 > 10$, then the feature space of a_2 is a subspace of the feature space of a_1 ; thus, condition a_2 is removed.

Procedure 1. COMBINERULESETS

Input: first ruleset (R_1), second ruleset (R_2), rule confidence threshold (α), and training set (D)

Output: combined ruleset R'

- 1: (Initialize a combined ruleset.) $R' \leftarrow \emptyset$
- 2: (Start a combination cycle.) **for each** ordered pair $(r_1, r_2) \in R_1 \times R_2$ **do**
- 3: (Create an empty new rule.) $r'.A \leftarrow \emptyset$ and $r'.y \leftarrow \text{NULL}$
- 4: (Merge conditions.) $r'.A \leftarrow r_1.A \cup r_2.A$
- 5: Remove subset conditions in $r'.A$
- 6: (Average the ensemble probability vectors.) $r'.p \leftarrow \mu(r_1.p, r_2.p)$
- 7: (Assign a predicted class.) $r'.y \leftarrow \arg\max_c r'.p(c)$
- 8: (Filter the new rule.) **if** $\text{cov}(r', D) > 0$ **and** $\text{conf}(r', D) > \alpha$ **then**
- 9: (Add the new rule.) $R' \leftarrow R' \cup \{r'\}$
- 10: (Return the combined ruleset.) **return** R'

In the second part, the class probability vectors for each source rule are aggregated by averaging both vectors $r_1.p$ and $r_2.p$ (see line 6). The predicted class of the new rule r' is obtained from the class with the highest probability in the probability vector meta-attribute $r'.p$ (see line 7). Subsequently, the rule heuristics are evaluated and, if the rule complies with the restrictions regarding coverage and confidence, the rule is added to the combined ruleset R' (see lines 8 and 9).

3.5. Ruleset simplification

The rules obtained in the growing stage are usually large, and tend to grow larger with every iteration, as more rules are combined and added to $R^\otimes$. Accordingly, the algorithm performs two simplification tasks: ruleset pruning and ruleset generalization.

3.5.1. Sequential covering pruning

The pruning process is based on the sequential covering strategy, a common rule learning strategy that learns a rule covering some samples of the training set, removes the samples covered by that rule, and recursively learns another rule covering some of the remaining samples until all samples are covered [10]. The function SEQUENTIALCOVERINGPRUNING has a main cycle that starts by computing the rule heuristics of R (see line 4) and sorting the rules in R in descending order, according to the rule accuracy and rule coverage (see line 5). Next, a nested cycle

Table 4

Comparison of F-measures ($\mu \pm \sigma$) among ensemble simplification algorithms for GBM in 3×10 -fold CV. A symbol marking indicates the result of the Wilcoxon signed-rank test, i.e., $\uparrow$: better, $\downarrow$: worse, and no mark: not statistically different (p-value $> .05$).

Dataset	GBM	GBM-RFIT	GBM-IT	GBM-DFT	RF-FBT	GBM-RC$\pm$	BSTR	DT
australian	84.18 $\pm$ 3.65	84.25 $\pm$ 4.00	26.25 $\pm$ 26.02	83.14 $\pm$ 4.47	83.74 $\pm$ 5.57	83.56 $\pm$ 3.36	84.94 $\pm$ 2.58	84.66 $\pm$ 2.99
bands	78.33 $\pm$ 5.04	77.23 $\pm$ 2.43	$\downarrow$ 78.88 $\pm$ 5.03	73.50 $\pm$ 4.63 $\downarrow$	71.86 $\pm$ 7.34 $\downarrow$	75.63 $\pm$ 4.56 $\downarrow$	74.33 $\pm$ 7.25	70.49 $\pm$ 5.95
banknote	99.86 $\pm$ 0.31	99.67 $\pm$ 0.55	96.04 $\pm$ 2.41 $\downarrow$	95.91 $\pm$ 2.04 $\downarrow$	98.91 $\pm$ 1.12 $\downarrow$	98.55 $\pm$ 1.13 $\downarrow$	90.48 $\pm$ 5.21	97.85 $\pm$ 1.50
biodeg	90.08 $\pm$ 2.62	88.61 $\pm$ 2.59 $\downarrow$	79.77 $\pm$ 1.63 $\downarrow$	84.54 $\pm$ 3.17 $\downarrow$	82.40 $\pm$ 5.33 $\downarrow$	89.16 $\pm$ 2.97 $\downarrow$	83.97 $\pm$ 1.46	86.47 $\pm$ 2.77
cardiotcg	97.47 $\pm$ 0.71	96.43 $\pm$ 1.13 $\downarrow$	93.21 $\pm$ 1.99 $\downarrow$	93.66 $\pm$ 1.32 $\downarrow$	89.26 $\pm$ 2.22 $\downarrow$	96.79 $\pm$ 0.78 $\downarrow$	94.22 $\pm$ 1.50	95.78 $\pm$ 1.17
credit	99.44 $\pm$ 2.14	99.14 $\pm$ 2.64	98.90 $\pm$ 3.94	99.44 $\pm$ 2.14	99.14 $\pm$ 2.64	99.44 $\pm$ 2.14	98.28 $\pm$ 3.52	99.14 $\pm$ 2.64
cryotherapy	88.66 $\pm$ 10.93	90.68 $\pm$ 10.48	87.82 $\pm$ 12.42	88.55 $\pm$ 10.88	89.48 $\pm$ 12.42	89.34 $\pm$ 10.23	89.86 $\pm$ 9.03	89.97 $\pm$ 11.14
divorce	97.88 $\pm$ 3.39	97.44 $\pm$ 3.20	95.58 $\pm$ 5.10	96.79 $\pm$ 3.78	98.10 $\pm$ 3.30	97.38 $\pm$ 3.34	96.56 $\pm$ 3.67	97.90 $\pm$ 3.44
ecoli0vs1	98.99 $\pm$ 1.57	98.61 $\pm$ 2.18	99.11 $\pm$ 1.74	99.09 $\pm$ 1.54	98.99 $\pm$ 1.57	98.99 $\pm$ 1.57	98.99 $\pm$ 1.57	98.99 $\pm$ 1.57
ecoli1	93.45 $\pm$ 3.54	93.07 $\pm$ 3.44	89.16 $\pm$ 2.95 $\downarrow$	93.06 $\pm$ 3.52	93.03 $\pm$ 3.80	93.48 $\pm$ 3.30	88.91 $\pm$ 2.02	93.71 $\pm$ 3.50
ecoli2	96.82 $\pm$ 1.65	96.61 $\pm$ 1.73	92.58 $\pm$ 2.12 $\downarrow$	95.75 $\pm$ 2.13 $\downarrow$	95.07 $\pm$ 3.38 $\downarrow$	95.99 $\pm$ 1.84 $\downarrow$	96.14 $\pm$ 2.21	96.25 $\pm$ 2.13
ecoli3	96.06 $\pm$ 1.61	96.30 $\pm$ 1.79	95.09 $\pm$ 3.21	95.90 $\pm$ 2.01	91.76 $\pm$ 4.00 $\downarrow$	95.68 $\pm$ 1.66	96.14 $\pm$ 1.47	95.39 $\pm$ 1.74
glass	96.10 $\pm$ 2.92	96.19 $\pm$ 3.31	93.57 $\pm$ 4.45	95.57 $\pm$ 3.35	85.92 $\pm$ 25.02	95.86 $\pm$ 3.03	93.80 $\pm$ 3.54	95.99 $\pm$ 3.21
glass0	88.28 $\pm$ 7.40	87.71 $\pm$ 7.84	81.03 $\pm$ 8.12 $\downarrow$	85.62 $\pm$ 8.14 $\downarrow$	82.99 $\pm$ 9.84 $\downarrow$	86.76 $\pm$ 6.87	82.91 $\pm$ 4.84	85.49 $\pm$ 7.76
glass1	86.13 $\pm$ 5.93	84.83 $\pm$ 8.90	78.29 $\pm$ 7.28 $\downarrow$	80.14 $\pm$ 7.26 $\downarrow$	80.61 $\pm$ 7.29 $\downarrow$	83.80 $\pm$ 8.22	82.47 $\pm$ 5.68	80.21 $\pm$ 9.16
glass6	97.42 $\pm$ 2.40	97.47 $\pm$ 2.72	96.93 $\pm$ 2.47	97.59 $\pm$ 2.50	92.89 $\pm$ 16.28	97.26 $\pm$ 2.25	98.15 $\pm$ 2.08	96.33 $\pm$ 3.07
haberman	83.65 $\pm$ 2.97	85.16 $\pm$ 2.56 $\uparrow$	84.46 $\pm$ 5.54	84.11 $\pm$ 3.90	80.42 $\pm$ 7.90 $\downarrow$	84.11 $\pm$ 2.64	85.51 $\pm$ 3.57	83.81 $\pm$ 4.96
heart	84.15 $\pm$ 6.88	81.62 $\pm$ 7.66 $\downarrow$	70.18 $\pm$ 8.19 $\downarrow$	81.49 $\pm$ 7.80 $\downarrow$	80.77 $\pm$ 6.59 $\downarrow$	82.70 $\pm$ 5.88	84.13 $\pm$ 5.76	81.09 $\pm$ 8.60
ionosphere	94.47 $\pm$ 3.00	94.37 $\pm$ 3.20	93.52 $\pm$ 3.55	93.20 $\pm$ 3.63	91.37 $\pm$ 4.39 $\downarrow$	92.05 $\pm$ 3.47 $\downarrow$	91.50 $\pm$ 3.83	91.54 $\pm$ 3.31
maintenance	75.83 $\pm$ 6.78	76.24 $\pm$ 6.49	68.08 $\pm$ 8.43 $\downarrow$	32.54 $\pm$ 11.65 $\downarrow$	14.77 $\pm$ 11.14	75.82 $\pm$ 5.84	32.97 $\pm$ 8.78	69.58 $\pm$ 5.93
mammographic	82.48 $\pm$ 5.15	82.01 $\pm$ 5.20	4.07 $\pm$ 15.72 $\downarrow$	81.95 $\pm$ 4.34	81.09 $\pm$ 4.93 $\downarrow$	81.87 $\pm$ 4.24	80.39 $\pm$ 4.81	82.40 $\pm$ 4.83
newthyroid1	98.64 $\pm$ 1.56	98.99 $\pm$ 1.35	98.09 $\pm$ 1.94	99.26 $\pm$ 1.24	97.05 $\pm$ 5.27	98.26 $\pm$ 1.56	99.00 $\pm$ 1.34	97.66 $\pm$ 2.13
newthyroid2	98.36 $\pm$ 1.83	98.83 $\pm$ 1.67	98.05 $\pm$ 2.35	98.27 $\pm$ 1.80	97.29 $\pm$ 2.47 $\downarrow$	98.09 $\pm$ 2.00	98.83 $\pm$ 1.80	97.58 $\pm$ 2.28
occupancy	99.60 $\pm$ 0.16	99.63 $\pm$ 0.16	94.36 $\pm$ 3.77 $\downarrow$	99.31 $\pm$ 0.25 $\downarrow$	98.58 $\pm$ 3.31 $\downarrow$	99.55 $\pm$ 0.17	99.28 $\pm$ 0.22	99.57 $\pm$ 0.16
pageblocks	98.59 $\pm$ 0.37	98.59 $\pm$ 0.40	96.01 $\pm$ 0.93 $\downarrow$	97.60 $\pm$ 0.42 $\downarrow$	95.68 $\pm$ 1.20 $\downarrow$	98.48 $\pm$ 0.36	97.55 $\pm$ 0.37	98.38 $\pm$ 0.43
pima	82.21 $\pm$ 2.64	78.94 $\pm$ 3.88 $\downarrow$	78.74 $\pm$ 4.10 $\downarrow$	81.20 $\pm$ 2.86 $\downarrow$	78.47 $\pm$ 8.51 $\downarrow$	82.48 $\pm$ 3.73	81.36 $\pm$ 2.72	80.54 $\pm$ 4.09
segment0	99.79 $\pm$ 0.20	99.86 $\pm$ 0.21 $\uparrow$	99.53 $\pm$ 0.32 $\downarrow$	99.54 $\pm$ 0.35 $\downarrow$	95.52 $\pm$ 2.61 $\downarrow$	99.66 $\pm$ 0.31 $\downarrow$	99.23 $\pm$ 0.54	99.54 $\pm$ 0.35
tic-tac-toe	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00	100.0 $\pm$ 0.00
vehicle2	98.83 $\pm$ 0.99	98.59 $\pm$ 1.19	95.90 $\pm$ 2.43 $\downarrow$	96.38 $\pm$ 2.00 $\downarrow$	88.58 $\pm$ 4.31 $\downarrow$	98.01 $\pm$ 1.50 $\downarrow$	91.06 $\pm$ 3.01	97.12 $\pm$ 1.59
vehicle3	86.38 $\pm$ 2.33	84.97 $\pm$ 2.65 $\downarrow$	86.11 $\pm$ 2.45	85.72 $\pm$ 3.08	83.91 $\pm$ 3.53 $\downarrow$	86.79 $\pm$ 1.62	84.25 $\pm$ 3.06	85.82 $\pm$ 1.75
wisconsin	97.25 $\pm$ 1.47	97.69 $\pm$ 1.57 $\uparrow$	94.91 $\pm$ 2.88 $\downarrow$	96.83 $\pm$ 1.68	94.87 $\pm$ 3.01 $\downarrow$	96.69 $\pm$ 1.88 $\downarrow$	96.55 $\pm$ 1.85	96.28 $\pm$ 2.03
yeast1	85.72 $\pm$ 1.96	83.29 $\pm$ 1.89 $\downarrow$	82.97 $\pm$ 2.41 $\downarrow$	83.87 $\pm$ 1.93 $\downarrow$	80.86 $\pm$ 4.01 $\downarrow$	85.14 $\pm$ 1.74	83.92 $\pm$ 1.63	83.13 $\pm$ 2.32
yeast3	97.22 $\pm$ 0.77	97.47 $\pm$ 0.87	94.29 $\pm$ 1.10 $\downarrow$	97.14 $\pm$ 0.94	96.57 $\pm$ 1.30 $\downarrow$	97.20 $\pm$ 1.12	94.30 $\pm$ 0.67	97.09 $\pm$ 1.13
Average	91.18 $\pm$ 2.84	92.14 $\pm$ 3.03 (improved)	85.50 $\pm$ 4.76	89.90 $\pm$ 3.36	87.57 $\pm$ 5.63	91.96 $\pm$ 2.89 (improved)	89.39 $\pm$ 3.08	91.08 $\pm$ 3.32

for implementing the sequential covering strategy begins to run by selecting the first rule with a rule accuracy greater than α . The nested cycle is broken if a rule with good accuracy and coverage is found (see lines 7 to 13). In the subsequent iteration of the main cycle, the ruleset heuristics are recomputed, and the ruleset is sorted again to match the remaining training data D' . The nested cycle starts again, and the procedure is repeated until a stopping criterion is met, that is, the rules in R are all removed, all of the training records in D are covered, or a good rule is not found. The sequential covering pruning strategy enhances the

performance of the algorithm in two ways. The execution time is reduced because the pruned rulesets are kept relatively small, and the general classification performance of the rules is improved.

Procedure 2. SEQUENTIALCOVERINGPRUNING

Input: combined unpruned ruleset (R), rule confidence threshold (α), and training set (D)

Output: pruned ruleset R'

- 1: (Initialize a pruned ruleset.) $R' \leftarrow \emptyset$
- 2: (Make a copy of training set.) $D' \leftarrow D$

(continued on next page)

(continued)

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3: (Start pruning cycle.) repeat
4:   (Compute rule heuristics.) Compute  $\text{cov}(r, D')$  and  $\text{conf}(r, D')$  for  $\forall r \in R$ 
5:   Sort rules in  $R$  based on  $\text{conf}$  and  $\text{cov}$ .
6:   (Create the flag triggered when a rule is found.)  $\text{found\_rule} \leftarrow \text{false}$ 
7:   (Iterate the rules in the input ruleset.) for  $r_i \in R$  do
8:     (Check the rule confidence threshold.) if  $\text{conf}(r_i, D') > \alpha$  then
9:       (Remove the samples covered by the rule.)  $D' \leftarrow D' \setminus \{x | x \in D' \wedge x \in \pi_{r_i}\}$ 
10:      (Remove the rule from original ruleset.)  $R \leftarrow R \setminus \{r_i\}$ 
11:      (Add the rule to the pruned ruleset.)  $R' \leftarrow R' \cup \{r_i\}$ 
12:      (Activate the found_rule flag.)  $\text{found\_rule} \leftarrow \text{true}$ 
13:      (Break this cycle.) break
14:   (Check stopping conditions.) until  $R = \emptyset$  or  $D' = \emptyset$  or  $\neg \text{found\_rule}$ 
15:   (Return the pruned ruleset.) return  $R'$ 

```

3.5.2. Ruleset generalization

The second task in the simplification stage of the algorithm is the ruleset generalization. In this process, the complexity of the rules is reduced to prevent the model from overfitting the training data. For this purpose, the *ERROR* function computes the statistical correction of the training error as an estimation of the generalization error of the rule r . This function assumes that the errors on r follow a binomial distribution and estimates the generalization error as the upper limit of the normal approximation of a binomial distribution of the training error e , as follows:

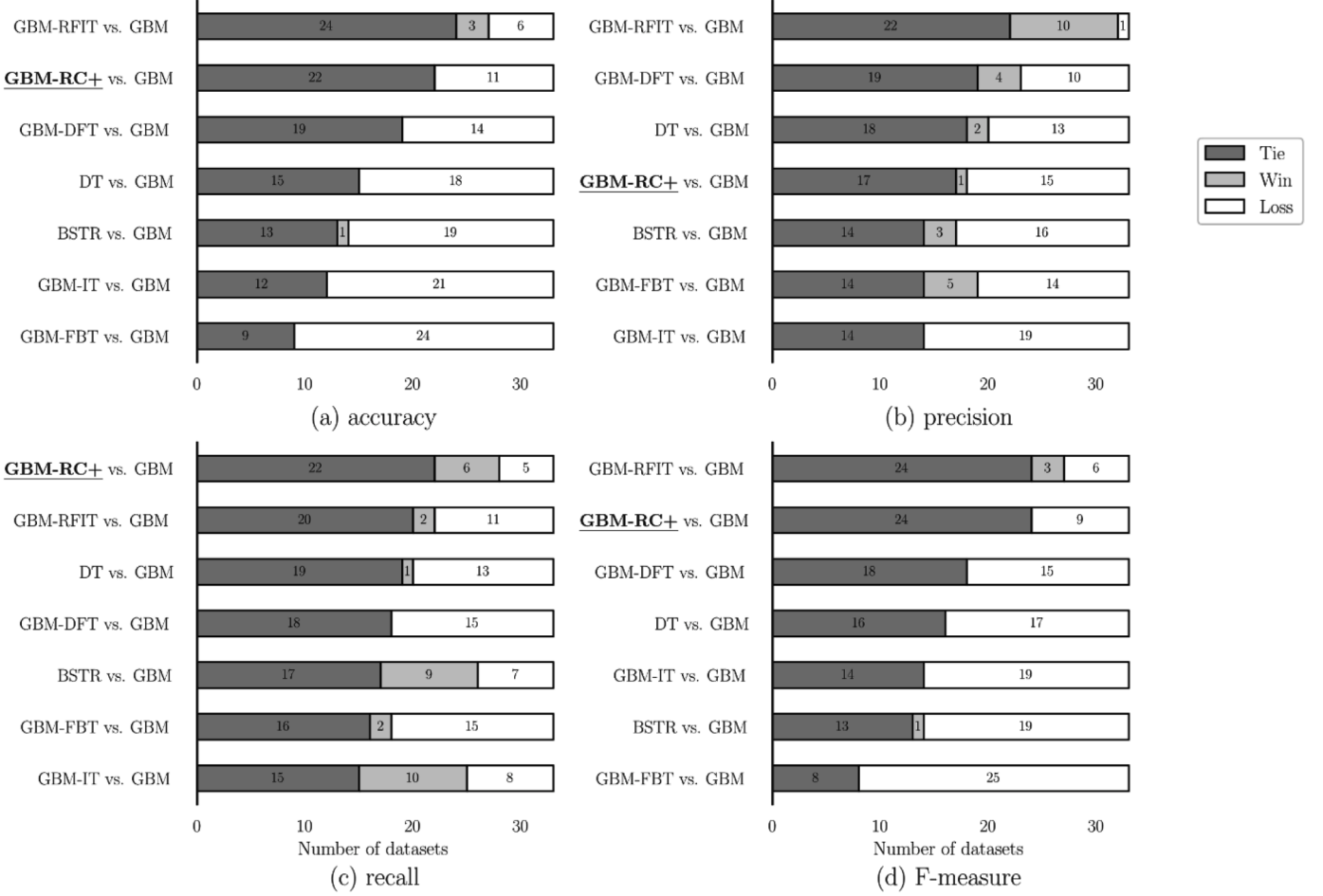


Fig. 2. Tie-win-loss record using the Wilcoxon signed-rank test among algorithms against original ensemble (GBM) over 33 datasets.

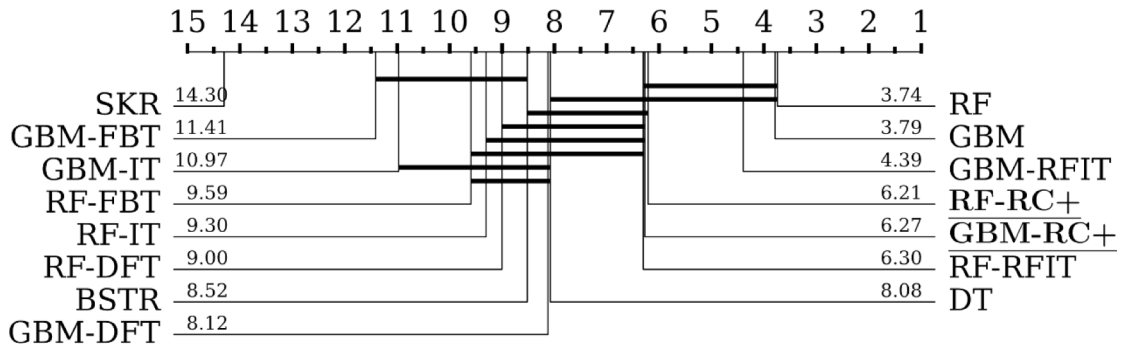


Fig. 3. Critical difference diagram of F-measure based on Wilcoxon-Holm post-hoc procedure (initial Friedman test result $F_f = 208.5$, $p \approx 0$).

Table 5

Comparison of F-measures ($\mu \pm \sigma$) between four tree ensembles and their RuleCOSI+ results in 3×10 -fold CV. A symbol marking indicates the result of the Wilcoxon signed-rank test, i.e., †: better, ‡: worse, and no mark: not statistically different (p-value > .05).

Dataset	BAG	<u>BAG-RC±</u>	XGB	<u>XGB-RC±</u>	CATB	<u>CATB-RC±</u>	LGBM	<u>LGBM-RC±</u>
australian	84.94 ± 3.55	83.40 ± 4.11 ↓	85.19 ± 4.26	82.96 ± 3.57 ↓	85.10 ± 3.90	83.70 ± 3.41 ↓	84.70 ± 3.29	83.70 ± 3.37
bands	78.92 ± 4.94	75.55 ± 4.36 ↓	79.16 ± 4.26	77.01 ± 1.91 ↓	81.10 ± 3.95	76.36 ± 3.39 ↓	77.55 ± 5.31	76.92 ± 1.42
banknote	98.74 ± 1.12	98.64 ± 1.20	99.46 ± 0.80	98.47 ± 1.74 ↓	99.76 ± 0.38	98.52 ± 1.49 ↓	99.70 ± 0.45	98.35 ± 1.91 ↓
biodeg	89.81 ± 2.63	87.79 ± 3.29 ↓	90.09 ± 2.62	87.53 ± 3.48 ↓	90.32 ± 2.64	87.79 ± 2.97 ↓	90.61 ± 2.46	88.14 ± 3.08 ↓
cardiotcg	96.50 ± 0.87	96.44 ± 0.98	97.34 ± 0.79	96.71 ± 0.83 ↓	97.22 ± 0.87	96.32 ± 1.01 ↓	97.43 ± 0.68	96.68 ± 1.00 ↓
credit	99.14 ± 2.64	99.14 ± 2.64	98.28 ± 3.52	98.28 ± 3.52	99.14 ± 2.64	99.14 ± 2.64	98.28 ± 3.52	98.28 ± 3.52
cryotherapy	91.22 ± 10.23	88.49 ± 10.04	91.08 ± 9.41	89.52 ± 8.10	91.38 ± 9.14	88.83 ± 9.87	89.28 ± 9.69	82.57 ± 14.28 ↓
divorce	97.24 ± 3.22	97.51 ± 3.52	97.44 ± 3.20	96.50 ± 3.83	97.44 ± 3.20	96.61 ± 3.87	97.24 ± 3.22	96.51 ± 4.05
ecoli0vs1	98.85 ± 1.95	99.10 ± 1.51	98.99 ± 1.57	98.99 ± 1.57	99.00 ± 1.76	98.85 ± 1.95	98.74 ± 1.90	98.62 ± 1.93
ecoli1	92.96 ± 2.55	93.34 ± 2.73	93.35 ± 3.03	92.68 ± 3.30	93.86 ± 2.93	91.35 ± 3.73 ↓	93.06 ± 3.19	92.32 ± 4.43
ecoli2	96.42 ± 1.90	96.08 ± 2.32	96.88 ± 1.77	96.52 ± 1.50	97.11 ± 1.59	96.14 ± 2.68	97.17 ± 1.74	96.60 ± 1.87 ↓
ecoli3	96.12 ± 1.66	95.95 ± 1.71	95.91 ± 1.93	95.14 ± 2.42	96.01 ± 1.57	95.13 ± 2.41 ↓	96.26 ± 1.48	95.28 ± 2.07 ↓
glass	96.38 ± 4.12	94.76 ± 3.51 ↓	95.57 ± 3.37	94.45 ± 3.42 ↓	95.97 ± 3.04	94.47 ± 3.95	96.31 ± 2.85	94.52 ± 3.58 ↓
glass0	89.38 ± 9.54	88.41 ± 7.36	89.02 ± 6.58	86.74 ± 10.00	90.83 ± 6.11	88.41 ± 6.70 ↓	88.48 ± 6.53	85.25 ± 9.47 ↓
glass1	84.78 ± 5.59	83.81 ± 7.02	86.63 ± 5.50	84.28 ± 8.20	88.27 ± 5.47	85.26 ± 8.01 ↓	86.58 ± 5.99	85.55 ± 6.87
glass6	97.13 ± 2.35	97.07 ± 2.48	98.33 ± 2.36	96.76 ± 2.45 ↓	98.23 ± 2.01	96.85 ± 3.25 ↓	98.25 ± 1.98	96.86 ± 2.74 ↓
haberman	83.91 ± 3.95	84.21 ± 2.18	83.50 ± 4.80	84.26 ± 2.73	84.29 ± 3.73	84.03 ± 3.10	82.85 ± 3.67	84.14 ± 2.97
heart	83.13 ± 6.53	80.36 ± 7.59 ↓	83.53 ± 6.27	80.00 ± 7.93 ↓	86.82 ± 6.12	81.76 ± 7.66 ↓	84.10 ± 5.30	81.06 ± 8.01 ↓
ionosphere	93.32 ± 3.19	93.00 ± 3.27	94.39 ± 2.91	91.87 ± 3.96 ↓	95.16 ± 3.04	92.40 ± 4.13 ↓	94.80 ± 3.15	93.28 ± 3.43 ↓
maintenance	63.93 ± 6.99	70.58 ± 8.90 †	76.64 ± 5.68	73.60 ± 7.65 ↓	75.65 ± 6.02	70.26 ± 6.32 ↓	76.30 ± 5.77	71.72 ± 6.32 ↓
mammographic	82.80 ± 4.93	82.39 ± 4.25	83.11 ± 4.72	82.38 ± 5.27	83.53 ± 4.65	81.88 ± 4.90 ↓	83.28 ± 4.74	82.83 ± 4.57
newthyroid1	98.62 ± 1.89	98.52 ± 1.61	98.46 ± 1.70	98.52 ± 1.77	99.37 ± 1.16	97.96 ± 1.94 ↓	98.53 ± 1.58	98.16 ± 2.21
newthyroid2	98.28 ± 2.04	98.26 ± 1.95	98.56 ± 1.82	98.14 ± 1.84	99.20 ± 1.70	97.95 ± 2.55 ↓	98.26 ± 1.84	98.42 ± 2.03
occupancy	99.59 ± 0.14	99.50 ± 0.20 ↓	99.60 ± 0.15	99.54 ± 0.18	99.63 ± 0.14	99.42 ± 0.25 ↓	99.58 ± 0.14	99.44 ± 0.21 ↓
pageblocks	98.56 ± 0.33	98.46 ± 0.36 ↓	98.66 ± 0.33	98.50 ± 0.41 ↓	98.66 ± 0.34	98.48 ± 0.28 ↓	98.63 ± 0.34	98.39 ± 0.37 ↓
pima	82.22 ± 3.19	81.38 ± 3.43 ↓	82.03 ± 3.43	80.44 ± 4.83 ↓	82.10 ± 3.03	80.55 ± 3.63	82.31 ± 2.79	80.58 ± 3.70 ↓
segment0	99.62 ± 0.27	99.64 ± 0.24	99.82 ± 0.19	99.66 ± 0.31 ↓	99.87 ± 0.18	99.73 ± 0.24 ↓	99.87 ± 0.18	99.70 ± 0.25 ↓
tic-tac-toe	100.0 ± 0.00	100.0 ± 0.00	100.0 ± 0.00	100.0 ± 0.00	100.0 ± 0.00	100.0 ± 0.00	100.00 ± 0.00	100.0 ± 0.00
vehicle2	97.81 ± 1.63	97.82 ± 1.44	98.91 ± 1.01	98.48 ± 1.31 ↓	98.88 ± 1.00	98.00 ± 1.50 ↓	99.12 ± 0.93	98.38 ± 1.28 ↓
vehicle3	86.83 ± 1.38	85.94 ± 2.16	86.56 ± 3.17	85.59 ± 1.69	86.67 ± 2.05	85.41 ± 2.08 ↓	87.08 ± 2.84	84.92 ± 2.24 ↓
wisconsin	97.14 ± 2.12	96.74 ± 1.87 ↓	97.13 ± 1.62	96.88 ± 1.65	97.55 ± 1.54	96.52 ± 2.51 ↓	97.13 ± 1.63	96.77 ± 1.74
yeast1	85.24 ± 1.78	84.46 ± 1.70 ↓	85.65 ± 2.12	83.54 ± 1.92 ↓	85.64 ± 1.90	83.69 ± 2.35 ↓	85.16 ± 2.23	83.53 ± 2.21 ↓
yeast3	97.12 ± 0.94	97.16 ± 0.95	97.20 ± 0.80	97.18 ± 1.04	97.27 ± 0.88	97.25 ± 0.97	97.31 ± 0.94	97.21 ± 1.04
Average	90.69 ± 2.99	91.63 ± 3.06 (improved)	91.30 ± 2.86	91.55 ± 3.16 (improved)	91.73 ± 2.66	91.49 ± 3.21	91.23 ± 2.76	91.35 ± 3.28 (improved)

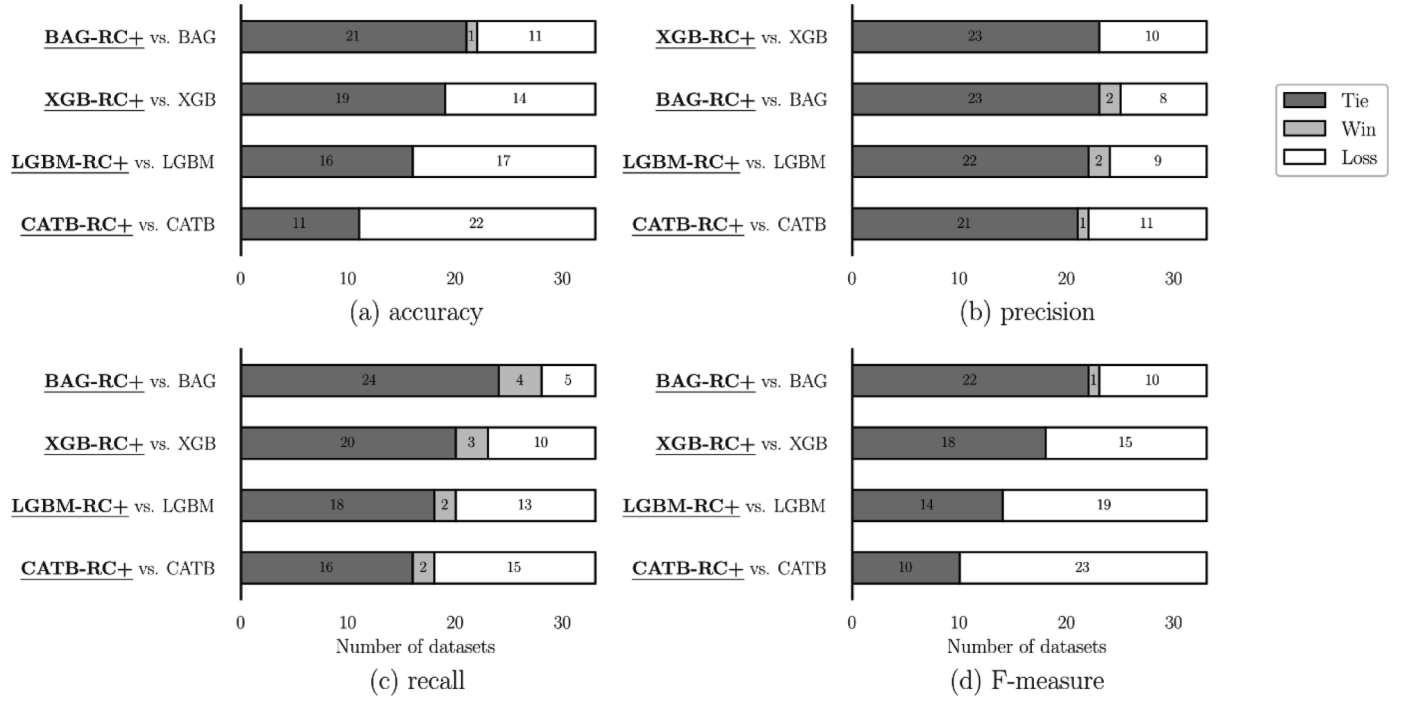


Fig. 4. Tie-win-loss record using the Wilcoxon signed-rank test between RuleCOSI+ and four original ensembles (BAG, XGB, CATB and LGBM) over 33 datasets.

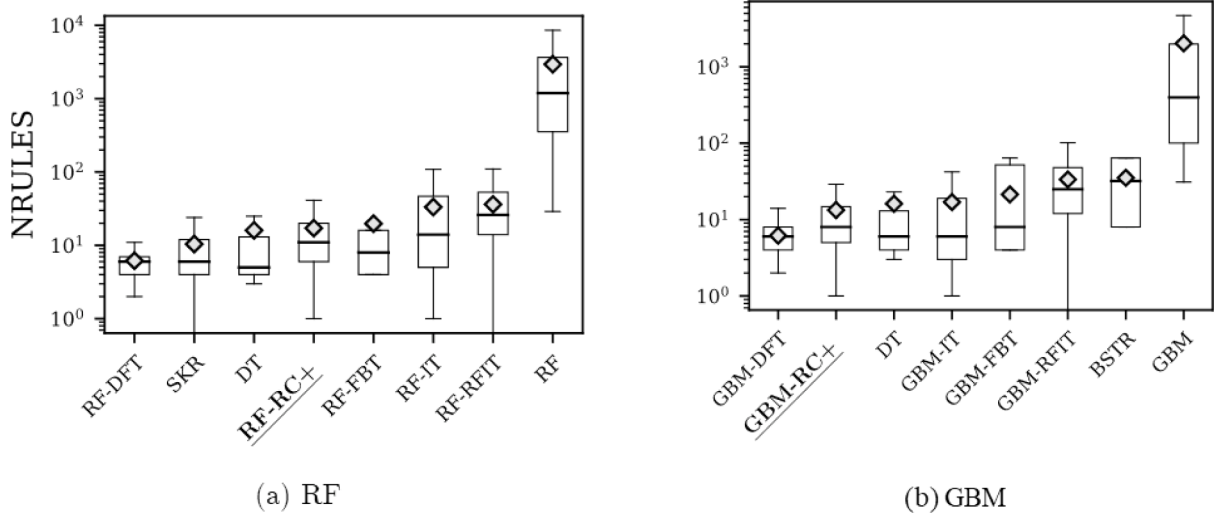


Fig. 5. Comparison of the number of rules (NRULES) among simplification algorithms in all the experiments for base ensembles: (a) RF and (b) GBM.

$$e_{upper}(n_r, e_r, c) = \frac{e_r + \frac{z_{c/2}^2}{2n_r} + z_{c/2} \sqrt{\frac{e_r(1-e_r)}{n_r} + \frac{z_{c/2}^2}{4n_r^2}}}{1 + \frac{z_{c/2}^2}{n_r}} \quad (2)$$

In the above, $n_r = |r.A|$ is the total number of training records covered by r , $e_r = 1 - \text{conf}(r)$ is the training error of r , c is the confidence level, and $z_{c/2}$ is the standardized value from the standard normal distribution [17]. The generalization process used by RuleCOSI+ is based on the C4.5 algorithm, therefore we used the same default value, that is $c = 25\%$, which yields a value of $z_{c/2} = 1.15$. The confidence level value of 25% is the default confidence level used in the C4.5 algorithm [18]. The confidence level c is a hyperparameter of the algorithm that controls

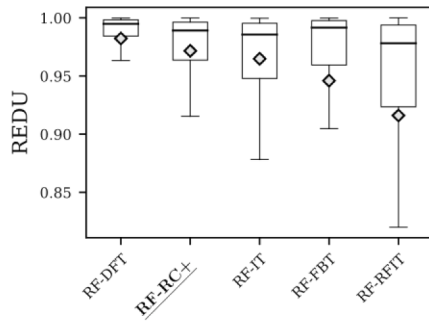
strength of the generalization applied to the simplified ruleset.

Procedure 3. GENERALIZERULESET

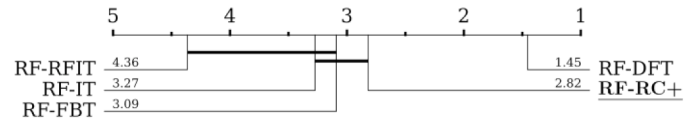
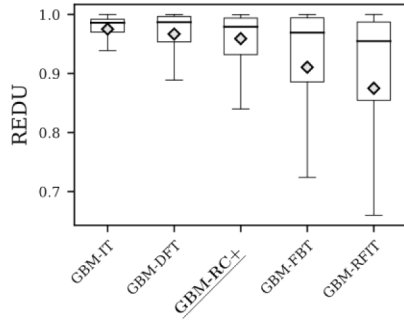
Input: ruleset (R), rule confidence threshold (α), rule coverage threshold (β), confidence level (c), and training set (D)
Output: simplified ruleset R

(Start individual rule processing.) **for each** $r \in R$ **do**
 2: (Compute the baseline generalization error.) $e^0 \leftarrow \text{Error}(r, c)$
 3: (Initialize the minimum error.) $e^* \leftarrow 0$
 4: (Start the generalization cycle.) **while** $e^* \leq e^0$ **and** $A! = \emptyset$ **do**
 5: (Find the condition that best improves the error.) $a^* \leftarrow \underset{a \in A}{\text{argminError}}(\{r|A = A \setminus \{a\}\}, c)$
 6: (Get the minimum error.) $e^* \leftarrow \text{Error}(\{r|A = A \setminus \{a^*\}\}, c)$
 7: (Check if the error was improved.) **if** $e^* \leq e^0$ **then**

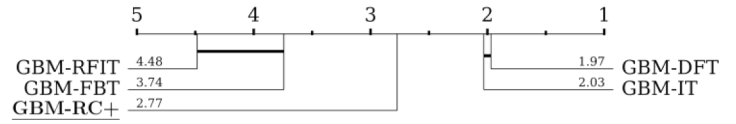
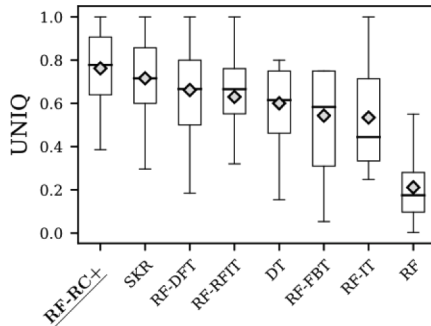
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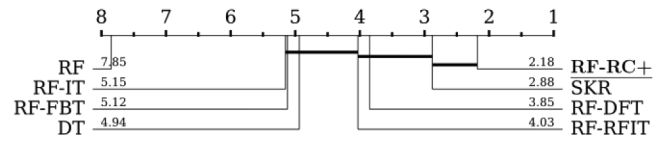
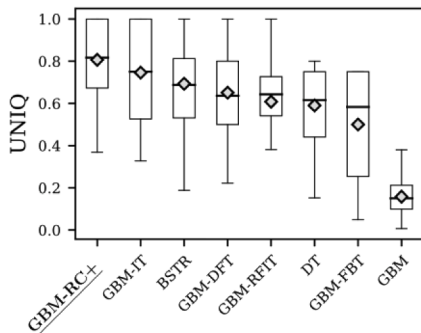
(a) REDU for RF

(b) Critical Difference Diagram for RF
(Initial Friedman test result $F_f = 57.6, p \approx 0$)

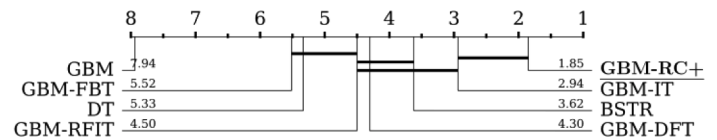
(c) REDU for GBM

(d) Critical Difference Diagram for GBM
(Initial Friedman test result $F_f = 63.6, p \approx 0$)**Fig. 6.** Comparison of the rule reduction rate (REDU) among simplification algorithms in all the experiments for base ensembles, RF and GBM.

(a) UNIQ for RF

(b) Critical Difference Diagram for RF
(Initial Friedman test result $F_f = 114.7, p \approx 0$)

(c) UNIQ for GBM

(d) Critical Difference Diagram for GBM
(Initial Friedman test result $F_f = 131.2, p \approx 0$)**Fig. 7.** Comparison of the antecedent uniqueness rate (uniq) among simplification algorithms in all the experiments for base ensembles, RF and GBM.

(continued)

8:	(Remove the best condition.) $A \leftarrow A / \{a^*\}$
9:	(Update the baseline error.) $\epsilon^0 \leftarrow \epsilon^*$
10:	(Initialize the minimum error.) $\epsilon^* \leftarrow 0$
11:	(Filter inaccurate or empty rules.) if $\text{cov}(r, D) > \beta$ and $\text{conf}(r, D) > \alpha$ and $A! = \emptyset$ then
12:	(Remove the rule.) $R \leftarrow R \setminus \{r\}$
13:	Sort rules in R based on conf and cov .
14:	(Return the simplified ruleset.) return R

Table 6

Classification and interpretability performance ($\mu \pm \sigma$) of simplification algorithms for RF and GBM, and baseline algorithms over 33 datasets.

Category	Algorithm	F-measure	NRULES	REDU	UNIQ
Ensemble simplification algorithms	RuleCOSI+ (RC+)	91.82 $\pm$ 8.8	15.3 $\pm$ 15.7	96.5 $\pm$ 4.3	78.4 $\pm$ 18.4
	RuleFit (RFIT)	91.81 $\pm$ 9.1	34.9 $\pm$ 30.5	89.4 $\pm$ 18.9	61.9 $\pm$ 20.8
	DefragTrees (DFT)	89.82 $\pm$ 13.0	6.2 $\pm$ 2.7	97.5 $\pm$ 3.8	65.6 $\pm$ 19.5
	Forest-based Tree (FBT)	88.39 $\pm$ 15.0	20.6 $\pm$ 23.6	92.3 $\pm$ 18.4	52.2 $\pm$ 23.8
	inTrees (IT)	87.08 $\pm$ 19.3	25.1 $\pm$ 36.3	97.0 $\pm$ 3.6	64.0 $\pm$ 26.4
	Decision Tree (DT)	91.08 $\pm$ 9.4	16.6 $\pm$ 31.0	N/A	56.9 $\pm$ 18.1
Baseline rule generation algorithms	BoostedRules (BSTR)	89.39 $\pm$ 12.7	35.2 $\pm$ 24.0	N/A	69.3 $\pm$ 19.7
	SkopeRules (SKR)	60.54 $\pm$ 39.8	10.4 $\pm$ 11.5	N/A	71.6 $\pm$ 20.2

Table 7

Classification and interpretability performance for the Australian credit approval case.

Category	Measure	GBM	GBM-RC+	GBM-RFIT	GBM-IT	GBM-DFT	GBM-FBT	BSTR	DT
Classification	Accuracy	86.96	89.86	82.61	56.52	89.86	88.41	86.96	86.96
	Precision	82.35	83.33	82.76	50.00	85.29	82.86	82.35	82.35
	Recall	90.32	96.77	77.42	13.33	93.55	93.55	90.32	90.32
	F-measure	86.15	89.55	80.00	21.05	89.23	87.88	86.15	86.15
Interpretability	NRULES	749	6	62	15	6	16	32	4
	REDU	N/A	99.20	91.72	98.00	99.20	97.86	N/A	N/A
	UNIQ	16.23	85.71	51.40	45.71	62.50	37.50	56.25	75.00

Table 8

Six rules extracted from GBM (749 rules in 50 trees) by using RuleCOSI+.

	COV	CONF	samples	r.A		r.y
r_1	41.87	97.31	[253,7]	$(A3 > 0.168) \wedge (A4 \leq 2.500) \wedge (A5 \leq 13.500) \wedge (A8 \leq 0.500)$	$\rightarrow$	[0]
r_2	16.59	98.06	[2,101]	$(A10 > 0.500) \wedge (A14 > 149.500) \wedge (A4 > 1.500) \wedge (A8 > 0.500)$	$\rightarrow$	[1]
r_3	2.42	100.00	[0,15]	$(A3 \leq 0.293) \wedge (A8 > 0.500)$	$\rightarrow$	[1]
r_4	1.61	100.00	[10,0]	$(A5 \leq 7.500) \wedge (A6 \leq 4.500) \wedge (A8 \leq 0.500)$	$\rightarrow$	[0]
r_5	0.32	100.00	[0,2]	$(A4 > 2.500)$	$\rightarrow$	[1]
r_d	37.20	65.37	[80,151]	()	$\rightarrow$	[1]

Table 9

Classification and interpretability performance for the breast cancer Wisconsin case.

Category	Measure	RF	RF-RC+	RF-RFIT	RF-IT	RF-DFT	RF-FBT	SKR	DT
Classification	Accuracy	100	98.55	100	95.65	98.55	89.86	95.65	97.10
	Precision	100	100	100	94.23	97.83	88.00	100	97.78
	Recall	100	97.78	100	100	100	97.78	93.33	97.78
	F-measure	100	98.88	100	97.03	98.90	92.63	96.55	97.78
Interpretability	NRULES	766	7	50	21	9	4	6	7
	REDU	N/A	99.09	93.47	97.26	98.83	99.48	N/A	N/A
	UNIQ	7.77	82.35	46.03	44.90	40.63	75.00	75.00	57.14

The generalization process function `GENERALIZERULESET` proceeds as follows. The rules are processed individually, starting by computing a baseline generalization error (see line 2). Then, the simplification procedure removes one condition from A at a time to find the condition a^* that minimizes the `ERROR` function (see line 5). If the minimized error improves over the baseline error, the condition a^* is removed, and the cycle starts again (see lines 7 to 10). This process is repeated until the `ERROR` function cannot be further improved, or until all of the conditions

Table 10

Seven rules extracted from RF (766 rules in 100 trees) by using RuleCOSI+.

	COV	CONF	samples	r.A		r.y
r_1	58.79	100	[0,361]	$(\text{BareNuclei} \leq 2.500) \wedge (\text{CellSize} \leq 3.500) \wedge (\text{NormalNucleoli} \leq 3.500)$	$\rightarrow$	[1]
r_2	19.87	99.18	[121,1]	$(\text{BareNuclei} > 2.500) \wedge (\text{ClumpThickness} > 6.500)$	$\rightarrow$	[0]
r_3	8.79	100	[54,0]	$(\text{BlandChromatin} > 3.500) \wedge (\text{CellSize} > 4.500) \wedge (\text{NormalNucleoli} > 2.500)$	$\rightarrow$	[0]
r_4	2.93	94.44	[1,17]	$(\text{BlandChromatin} \leq 2.500) \wedge (\text{ClumpThickness} \leq 8.500) \wedge (\text{NormalNucleoli} \leq 3.500)$	$\rightarrow$	[1]
r_5	1.79	100	[11,0]	$(\text{BareNuclei} > 8.500) \wedge (\text{BlandChromatin} > 3.500)$	$\rightarrow$	[0]
r_6	1.30	100	[0,8]	$(\text{BareNuclei} \leq 3.500) \wedge (\text{CellShape} \leq 2.500) \wedge (\text{ClumpThickness} \leq 8.500) \wedge (\text{MarginalAdhesion} \leq 7.000)$	$\rightarrow$	[1]
r_d	6.51	70.0	[28,12]	()	$\rightarrow$	[0]

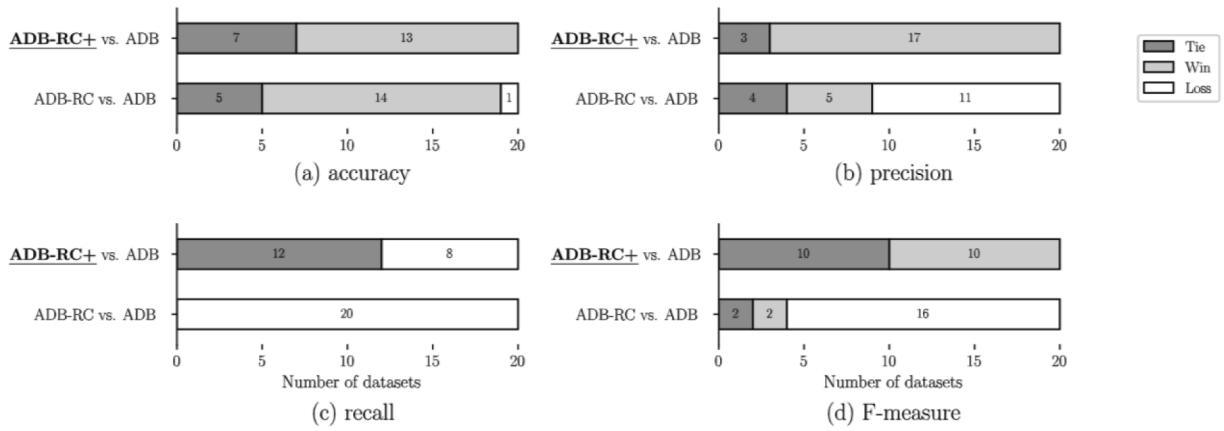


Fig. 8. Tie-win-loss record for the classification metrics of RC+ against original RC over 20 imbalanced datasets presented in [9].

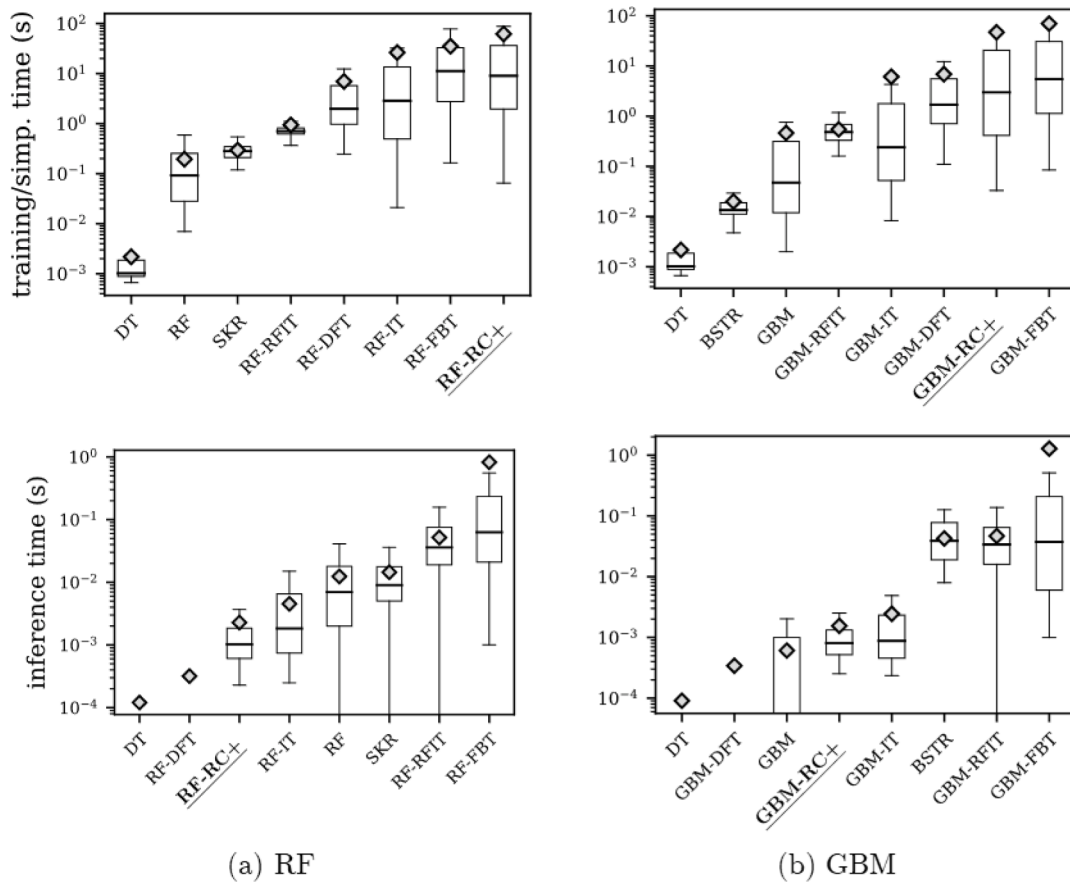


Fig. 9. Comparison of the training/simplification time and the inference time in all the experiments over 33 datasets for (a) RF and (b) GBM.

$a \in A$ are removed. The last step consists of removing the rules that do not comply with the rule accuracy and rule confidence thresholds (see lines 11 and 12).

3.6. Final steps: stopping criterion and default rule

The combination and simplification procedures are repeated until all of the rulesets are processed, or until the stopping criterion is met (see line 8 of the RuleCOSI+ algorithm). The criterion evaluates whether the performance of the simplified ruleset is perfect. Generally, a perfect

performance measure has a value of one. For example, in our experiments, we used the F-measure as performance measure; thus, if R^* had a $F\text{-measure} = 1$, the combination process was stopped.

The last step in the algorithm consists of generating a default rule r_d . The set of conditions A of r_d is empty, and the predicted class is decided based on the majority class of the uncovered instances in D . If all instances are covered by the existing rules, the predicted class is determined as the prior majority class of the training set.

3.7. Complexity time and implementation details

We presented RuleCOSI+ algorithm in detail in the previous sub-sections. The algorithm heavily depends on the computation of the rule heuristic functions cov and conf based on the training set D . The greater the number of combinations, the more computations of the heuristics that must be performed. The time complexity of computing rule heuristics is $O(n)$ where n is the number of records in the training set. The algorithm is composed of three functions, the CombineRulesets function time complexity is $O(n4^d)$, where d is the maximum depth of each tree; the SequentialCoveringPruning function time complexity is also $O(n4^d)$; and the GeneralizeRuleset function time complexity is $O(n8^d)$. The total time complexity of RuleCOSI+ is $O((K-1)n(2^{2d+1} + 8^d)) = O(Kn8^d)$ where K is the number of trees in the ensemble. The algorithm is linear with respect of n and exponential with respect of d . The main factor affecting the execution time of the proposed algorithm is the maximum depth of the trees because deep trees lead to longer and more specialized rules. Therefore, for the proposed algorithm to have practical value, the rule heuristics computations must be efficient. Here, we present two specific implementation details that we used in our experiments.

The first implementation detail is to keep a record of the results from past rule combinations. Two sets were used for this purpose. The first was a “good combinations set” $\mathbb{G}$ for storing all of the rule combinations complying with the rule confidence and rule coverage restrictions, this is, $\mathbb{G} = \{r \mid \text{cov}(r) > 0 \wedge \text{conf}(r) > \alpha\}$. The second was a “bad combinations set” $\mathbb{B}$ for storing the rule combinations failing to comply with the rule confidence and rule coverage restrictions, i.e., $\mathbb{B} = \{r \mid \text{cov}(r) = 0 \vee \text{conf}(r) \leq \alpha\}$. Both sets help to avoid computing the rule heuristics more than once for the same combination of rules.

The second implementation detail was computing the coverage of all the conditions extracted from the tree ensemble and storing them in binary-valued arrays of bits¹. Each array was of size n , and was constructed based on the training set D . If an instance x_i was covered by condition a , the position i of the array obtained a *true* value, and if not, it obtained a *false* value. Using these arrays, every time a combination is performed, the heuristics are efficiently computed using set operations on the arrays of the conditions in the new combined rule condition set.

The previous implementation details contributed to reducing the execution times of RuleCOSI+ and enabled the algorithm to handle large ensembles. In our experiments, we used ensembles with up to 500 trees; nevertheless, the algorithm can handle larger tree ensembles restricted only by the memory used for storing large datasets.

4. Measuring tree ensemble interpretability

To compare RuleCOSI+ with other similar ensemble simplification methods, we define specialized measures that can fairly compare the respectively interpretability of models such as tree ensembles, rulesets, or DTs. For simplicity, we measure the complexity of a tree ensemble based on its size. The size of H can be taken as the number of trees conforming to the ensemble, or it could be considered as the sum of the total number of terminal nodes t on each tree h_k . Instead, we use a function ψ for transforming a tree h into a ruleset, i.e., $R = \psi(h)$. Then, we use the *total number of rules* (denoted by the function Nrules) extracted from each tree $h_k \in H$ as the size of the ensemble, as follows:

$$\text{Nrules}(H) = \sum_{k=1}^K \text{Nrules}(h_k) = \sum_{k=1}^K |\psi(h_k)| \quad (3)$$

Indeed, the number of rules is one of the most commonly used complexity measures for ensemble rule extraction methods [5–7,9].

A more intuitive way to compare the complexity of an ensemble

model H and ruleset R is to directly compare the sizes of the two models in terms of the function Nrules . For this purpose, we introduce a new measure for quantifying the percentage of the reduction in size between two tree-like models. The measure is called the *rule reduction rate* and is denoted by redu . The equation for this measure is as follows:

$$\text{redu}(R|H) = 1 - \frac{\text{Nrules}(R)}{\text{Nrules}(H)} \quad (4)$$

Additionally, a new interpretability measure is introduced to compare decision-tree-like models. The purpose of this measure is to quantify the amount of non-redundant information contained in the model. The rationale behind this measure is that if a model contains repeated information, it is more difficult to interpret. We define the set $\mathbb{A} = \{a \mid a \in r_i, \forall r_i \in R\}$, which has all the conditions belonging to all the rules of a ruleset R without duplicates, and the multiset $\mathbb{A}^*$, which has the same definition as $\mathbb{A}$, but allows duplicates. The *antecedent uniqueness rate* or uniq measure of a rule extraction G , which can be h , H or R , is defined as follows:

$$\text{uniq}(G) = \frac{|\mathbb{A}|}{|\mathbb{A}^*|} \quad (5)$$

Here, $\mathbb{A}$ is the set, and $\mathbb{A}^*$ is the multiset of conditions from the extracted ruleset $R = \psi(G)$. The measure ranges from 0 to 1. The higher the number, the simpler and easier it is to understand the model, as it does not contain redundant conditions.

With this complexity-measuring framework, we present the experiments conducted to validate the proposed rule extraction algorithm.

5. Experimental design

In this section, we present the design of the empirical experiments conducted to prove the applicability of the proposed method. The proposed algorithm was implemented as a Python library², based on the scikit-learn package v1.0.2 [19].

5.1. Datasets

We used 33 binary class versions of several publicly available datasets from the UCI Machine Learning Repository. Some of the original datasets are for multiclass classification; therefore, the data were adjusted so that in some cases the union of one or more classes conformed to one class, and in other cases, the union became the other class. The properties of all the datasets used, such as the majority and minority class information, imbalanced ratio (IR), number of instances, and number of attributes, are presented in Table 1.

5.2. Tree ensemble and simplification algorithms

Six different tree ensemble algorithms were used in the experiments, including two independent algorithms, bagging of trees (BAG) [20] and random forest (RF) [21], and four different implementations of the GBT, the original gradient boosting machine (GBM) [22], XGBoost (XGB) [23], LightGBM (LGBM), [24] and CatBoost (CATB) [25].

We compared the results of RuleCOSI+ (denoted by RC+ in tables and figures) with those of four simplification models and with three different types of single interpretable models. For the simplification methods, we performed the comparisons using RF and GBM algorithms as base ensembles.

¹ We specifically used the Python built-in integer type and NumPy binary-valued uint8 arrays.

² The implementation is publicly available in the following repository <https://github.com/jobregon1212/rulecosi>

- *RuleFit* (RFIT) is the classic rule ensemble method presented by Friedman and Popescu [4]. The Python implementation provided by the *imodels* package³ was used.
- *inTrees* (IT) was developed by Deng [5]. The R package implemented by the author⁴ was used.
- *DefragTrees* (DFT) is the Bayesian model selection approach presented Hara and Hayashi [6]. The Python implementation provided by the authors⁵ was used.
- *Forest-based Trees* (FBT) was first proposed for independent tree ensembles by Sagi and Rokach [7], and then was extended to dependent tree ensembles [8]. The Python implementations provided by the authors⁶ was used. This method was included to compare RuleCOSI+ with other post-hoc simplification methods that do not output a ruleset, and instead output a decision tree.
- *Decision trees* (DT) were tested because it is a good way to compare a simple interpretable model with the results from simplification models.
- *SkopeRules* (SKR) works by extracting rules from tree ensembles and then removing similar or duplicated rules based on rule support. The Python implementation provided by the scikit-learn-contrib community⁷ was used.
- *BoostedRules* (BSTR) is a rule boosting method that sequentially fits rules following the AdaBoost algorithm [26]. The Python implementation used is also provided by the *imodels* package.

5.3. Model evaluation and hyperparameter optimization

The experiments were performed using a stratified 3×10 fold cross-validation (CV) scheme, which involves a 10-fold CV repeated three times. For each fold, a 5-fold CV grid search was used to optimize the hyperparameters of the tree ensemble. For this study, we only considered two hyperparameters for the tree ensembles: the number of trees, $n_{\text{trees}} \in \{10, 25, 50, 100, 250, 500\}$, and the maximum depth for each tree, $d \in \{2, 3, 4, 5, 6\}$. Having the optimal ensemble parameters, an additional 3-fold CV grid search was used to optimize the hyperparameters of all the simplification and interpretable baseline methods. The search space of the hyperparameters for each algorithm is presented in Table 2.

5.4. Statistical tests

The experiments were validated based on two statistical tests. First, an individual dataset pairwise comparison was performed using the Wilcoxon signed-rank test over 30 pair results of each dataset as recommended by Demšar [27]. The null hypothesis assumption was that for a randomly drawn training set of a fixed size, the original ensemble and simplified result have the same classification performance on a randomly drawn test sample; in other words, the mean performances of the original ensemble and combined ruleset are equivalent.

Second, we compared the classifiers using multiple datasets. The process started by applying a Friedman test to reject the null hypothesis (that there was a statistical difference between the compared algorithms) [27]. After that, the Wilcoxon signed-rank test with the Holm's alpha correction post-hoc procedure was applied, as suggested by Benavoli, Corani, and Mangili [28].

6. Experiment results

The experimental results and their analyses are presented in this

section. There are two points to be evaluated.

- 1 To assess whether the classification performance of the simplified ruleset is approximately equal to that of the original ensemble. This is evaluated using statistical tests.
- 2 To confirm that the proposed method decreases the complexity of the original tree ensembles. This is evaluated using the interpretability measures introduced in Section 4.

6.1. Classification results

The classification performance of all the models was assessed using four main measures: accuracy, precision, recall, and F-measure. However, we focus our analysis using the last one because it represents a good balance between precision and recall. As most post-hoc classification methods only work with the RF and GBM, our classification results are divided into three parts as follows:

- 1 The classification results of the RF are presented in Table 3 and summarized in Figure 1 for the tie-loss-win record pairwise comparison and Figure 3 for the multiple-dataset comparisons.
- 2 The classification results of the GBM are presented in Table 4 and summarized in Figure 2 for the tie-loss vs. win record pairwise comparison and Figure 3 for the multiple dataset comparisons.
- 3 The classification results of the remaining algorithms (BAG, XGB, CATB, and LGBM) are presented in Table 5 and summarized in Figure 4.

The results of the experiments using an independent ensemble (RF) are presented in Table 3 and Figure 1. RF-RC+ could maintain or improve the F-measure in 23 datasets (69.7%), whereas the results of RF-RFIT and RF-IT algorithms were slightly worse in maintaining the F-measure in 20 and 18 datasets respectively (60.6% and 54.5%). Figure 1 shows that the proposed method has a better recall than precision compared with other algorithms, which means that it can better detect positive examples in the data. However, it tends to make slightly more mistakes when doing so. RF-RFIT algorithm obtained good results, but, as it is discussed in the next subsection, it tends to produce more rules compared to other simplification methods, which reduces its interpretability. Other simplification algorithms —RF-FBT and DFT— and single models —SKR and DT— obtained worse results compared with simplification algorithms.

The results from the experiments using a dependent algorithm (GBM) are presented in Table 4 and Figure 2. GBM-RC+ could maintain the F-measure in 24 datasets (72.7%), whereas other simplification algorithms obtained worse results, except GBM-RFIT with 27 datasets (81.8%). In the case of GBM, the proposed method also shown better recall than precision, GBM-RC+ maintained or improved the recall in 28 datasets (84.8%) but had difficulty maintaining the precision metric with only 18 datasets (54.5%). GBM-DFT performed better than RF-DFT. Single benchmark models, namely, BSTR and DT, and the remaining simplification algorithms —GBM-IT and GBM-FBT—obtained worse results compared with GBM-RFIT and GBM-RC+.

The statistical comparison results of all the models for the RF and GBM are presented in Figure 3. The number appearing beside each algorithm represents the average rank obtained over 33 datasets in the experiments. Following the statistical procedure presented in Section 5.4, the Friedman test rejected the null hypothesis, that is, all the algorithms had the same performance in terms of the F-measure ($F_f = 208.47$, $p \approx 0$).

Subsequently, the post-hoc Wilcoxon signed-rank test with Holm's alpha correction was performed. In Figure 3, a thick horizontal line connects algorithms which do not have statistically significant difference in classification performance based on F-measure. The original ensembles, RF and GBM, obtained the first positions and exhibited

³ <https://github.com/csinva/imodels>

⁴ <https://cran.r-project.org/package=inTrees>

⁵ <https://github.com/sato9hara/defragTrees>

⁶ <https://github.com/sagyome/XGBoostTreeApproximator>

⁷ <https://github.com/scikit-learn-contrib/skope-rules>

similar performances. GBM-RFIT, RF-RC+, GBM-RC+, RF-RFIT, DT and GBM-DFT have lines connected at least with one ensemble, which indicates that the sizes of tree ensembles can be reduced without significantly affecting their performance with RC+ and RFIT. The left side of Figure 3 shows the models that tend to appear in the lower half of the experiments; IT and FBT with both base ensembles and RF-DFT seemed to have performed poorly along with BSTR, when compared with the base ensembles. However, many middle-ranked methods had a similar performance. Finally, SKR performed poorly in all experiments.

Table 5 presents the classification results from the experiments using BAG, XGB, CATB, and LGBM as the base ensembles. BAG-RC+ performed well and achieved a similar performance relative to that of the original ensembles in 23 datasets (69.7%). XGB-RC+ had a modest performance with 18 datasets (54.5%). The application of RuleCOSI+ algorithm to more advanced gradient boosting algorithms such as LGBM and CATB resulted in the worst performance of the proposed method, with only 14 and 10 datasets (42.4% and 30.3%). These results are depicted in Figure 4.

6.2. Interpretability results

The interpretability measures used for comparing the post-hoc simplification methods are NRULES, REDU, and UNIQ, as introduced in Section 4.

First, the results from one of the most common interpretability metrics, the number of rules in the models (NRULES), are presented. A boxplot comparing the results obtained in all the experiments for the base ensembles RF and GBM is presented in Figure 5. Notably, the number of rules obtained using all the simplification methods is significantly reduced relative to those from the original ensembles. The GBM ensembles tend to generate fewer rules as compared to those of the RF to achieve the same level of performance. We can observe three groups of models: those that generate less than 10 rules—RF-DFT, GBM-DFT, DT, SRK and GBM-RC+—, those that generate between 10 and 20 rules in average—RF-RC+, RF-FBT, GBM-IT and GBM-FBT— and those that generates more than 20 rules—RF-IT, RF-RFIT, GBM-RFIT and BSTR. This is one the reasons why RFIT can maintain the classification performance in more datasets as compared to RuleCOSI+.

The next step is to compare the rule reduction rate or REDU measure for all the post-hoc simplification methods. The REDU results for all the experiments are summarized using several graphs in Figure 6. As RF models tend to have more rules (see Figure 5(a)), the REDU measure is consequently higher for these models, as shown in Figure 6(a). RF-DFT and GBM-DFT models have more consistent results since the maximum number of rules is a hyperparameter that can be controlled. Models simplified using IT and RuleCOSI+ show similar rule reduction rates. Finally, RFIT and FBT have more variability in the rule reduction that can be achieved.

A statistical comparison of the REDU measure was performed, and the results are presented in Figure 6(b) for the RF simplified models and in Figure 6(d) for the GBM simplified models. For the RF simplified models, RF-DFT was the best in reducing the size of the models. This outcome was expected because one of the hyperparameters of DFT is the maximum number of rules, as shown in Table 3. RC+ ranked second, while RF-IT and RF-FBT shown similar behavior. RF-RFIT ranked last. For the GBM base ensembles, GBM-DFT and GBM-IT ranked first and GBM-RC+ ranked second. GBM-FBT and GBM-RFIT ranked last.

Finally, the original ensembles of all the simplified models and the interpretable baseline models are compared in terms of the amount of

non-redundant information in them. For this purpose, we used the UNIQ measure. Figure 7 presents several graphs summarizing the results of all the experiments for the RF and GBM.

RF and GBM ensembles have very low unique values, between 16% and 21% of non-redundant information, as shown in Figure 7(a) and Figure 7(c), respectively. This outcome is expected from tree ensembles, as many branches are repeated among trees. RF-IT models have very low UNIQ scores possibly because of the diversity of rules extracted from RF ensembles. In the middle, RF-FBT, DT, RF-RFIT and GBM-RFIT have approximately 60% of non-redundant information, followed by the BSTR, SKR, and GBM-IT models with approximately 70% of non-redundant information. At the top of both boxplots, the proposed method RuleCOSI+ has approximately 80% of the non-redundant information in the generated rules. This is also supported by the statistical tests performed using the UNIQ measure. The results from the statistical tests are presented in Figure 7(b) for RF and in Figure 7(d) for the GBM. GBM-RC+ and GBM-IT outperformed all the other models, while RF-RC+ is at the top, with the interpretable baseline models generated with SKR.

6.3. Summary of comparative experiments

The summary of all the classification and interpretability results presented in the previous subsections is given in Table 6. The best simplified classifiers were obtained by RFIT and RC+ with an F-measure of 91.8, and RC+ had less variability in the results. However, as it can be seen RFIT generated on average 34.9 rules compared with RC+ that generated only 15.3 rules on average. Some algorithms such as DFT generated less rules, but the classification performance measure was lower compared with RC+ and RFIT. Additionally, the proposed method outperformed all other methods in terms of generating models with non-redundant information, making the models easier to understand. In summary, RuleCOSI+ is a post-hoc ensemble simplification algorithm that generates small and accurate rulesets with non-redundant information, in contrast with other existing methods.

6.4. Practical examples

To prove the practical applicability of RuleCOSI+, we present two examples using two datasets from our experiments. The models presented were extracted from one of the folds of the 3×10 fold CV experiments.

6.4.1. Australian credit approval data

The first case study is in the financial domain, specifically in the credit risk management field. The use of machine learning predictive models in this field is common, but it poses questions about data security and fair lending. Certainly, “the financial industry is highly regulated and loan issuers are required by law to make fair decisions” [29]. Therefore, it is important to understand the decision mechanisms of accurate models, such as tree ensembles. The dataset consists of 14 anonymized features of credit card applications, and the goal is to classify the applicants as approved or not approved.

The experiment started by training a GBM model with 50 trees with a maximum depth of 4. The result is too big to be shown here, so it is not included. The summary of the classification and interpretability measures is presented in Table 7. The GBM model has an F-measure of 86.15 and size of 749 rules, which is the number of leaf nodes in all the trees conforming the ensemble. It has only 16.23% of non-redundant

information. We extracted rules from the ensemble using the proposed model and other simplification benchmark algorithms. In addition, two single interpretable models were trained, namely, BSTR and DT. In this specific study case, RuleCOSI+ outperformed other algorithms in terms of the classification and interpretability metrics. On the one hand, the GBM-FBT model has a slightly lesser F-measure (87.88) but a bigger set of rules (16) and only 37.5% of non-redundant information. On the other hand, the DT model has a smaller number of rules (4) but a low F-measure of 86.15. GBM-DFT obtained the same number of rules but with only 62.50% of non-redundant information, in contrast with 85.71% obtained by RuleCOSI+. This shows that our proposed model provides a better balance between accurate models and interpretable models compared to other similar methods.

The combined and simplified set of rules obtained by GBM-RuleCOSI+ is presented in Table 8. The ruleset has only six rules. The first three rules cover approximately 60% of the cases with a high confidence, between 97.31% and 100%. The remaining 40% cases are covered by the last two rules and the default rule, which has a 65.37% confidence. The rules are highly interpretable and provide a global interpretable model to understand the decisions derived from a GBM ensemble.

6.4.2. Breast cancer Wisconsin data

The second case study is in the medical domain. Freitas [30] emphasized that medical users of predictive models valued good explanations. Therefore, interpretable models are desirable for providing a good diagnosis to patients. The dataset consists of nine features extracted from digitized images of a fine-needle aspirate of a breast mass. The features describe the cell nuclei contained in the image. The goal of the classification task is to determine if the tumor is benign or malignant.

The experiment started by training an RF model with 100 trees with a maximum depth of 3. The result is too big to be shown here, so it is not included. The summary of the classification and interpretability measures is presented in Table 9. The RF model has an F-measure of 100 and size of 766 rules. It has only 7.77% of non-redundant information. We extracted rules from the ensemble in a similar fashion as in the previous case study, except that an SKR model was trained instead of a BSTR.

In this case study, RF-RuleCOSI+ represents a good balance between classification and interpretability. The best performing classifiers are RF and RF-RFIT, with perfect classification score. However, both classifiers have many rules (766 and 50 respectively). In terms of interpretability, RF-FBT obtained just four rules, but the F-measure is low (92.63). RF-RC+ generated the same rules as DT (7) but with better classification performance—98.88 compared to 97.78—and with fewer redundant information—82.35% compared to 57.14%.

The combined and simplified set of rules obtained by RF-RuleCOSI+ is presented in Table 10. The ruleset has seven rules. The first four rules cover approximately 90% of the cases with high confidence. Other rules— r_5 and r_6 —cover fewer examples and can be analyzed by domain experts to understand the relationship between rules and the predicted response. Moreover, in this case, the rules provided by RuleCOSI+ are global explanations to understand rules that cover common cases and rare cases in the data.

7. Discussion

The algorithm presented in this work is an improvement of the methods proposed in the literature. It is a practical machine learning approach because it has a good classification performance and produces small rulesets with low redundant information, making the models easy

to interpret. We discuss a few related topics in the following subsections before concluding the paper.

7.1. Dependent vs. independent tree ensembles

Based on the obtained results, the algorithm has slightly different results with RF as compared to GBM. The difference lies in the nature of each ensemble learning framework. In the case of RF, because of the random nature of each tree, the proposed method can explore more diverse feature space subregions and obtain good results, but with more rules—16 compared to 12 rules on average of the GBM. In the case of GBM, the constructed trees tend to be shallow, so the number of rules is reduced. In addition, applying the proposed method to GBM variants, such as the LGBM and CATB, yielded poor results when comparing the simplified models against original ensembles. This could be because of the improvements in the gradient boosting algorithm, such as the use of regularization methods when training each tree. Additionally, we suspect that the proposed algorithm obtained poor results with CATB because the base predictors used in that algorithm are oblivious decision trees, which are also called decision tables [25].

7.2. Heuristic search

The heuristic search used in the proposed method allows for the exploration of diverse combinations of regions of the feature space provided by the original tree ensembles. Consequently, the rules generated by RuleCOSI+ are unique in the sense that they are not equivalent to any of the trees inside the ensemble, because the simplified rules are formed with a combination of regions on the feature space that reasonably approximates the predictions of the original tree ensemble while significantly reducing the ensemble size, as shown in Figure 3.

7.3. Interpretability of the result

Decision lists. The outputs of the proposed algorithm are decision lists with interpretability advantages relative to other types of transparent models.

- **Unordered rulesets.** Decision lists are easier to interpret than ordered rulesets in two ways: (a) an unordered ruleset should evaluate all of the rules when making predictions, whereas decision lists need only to evaluate the rules until one is fired; and (b) ordered rules should have an aggregation mechanism in the case where one example is covered by more than one rule (this is one of the disadvantages of using tree ensembles).
- **Decision trees.** Decision trees represent non-overlapping regions of training data, which affect the depth of a tree when complex regions are described by the tree structure. In contrast, decision lists allow for the representation of overlapping regions of the feature space in different rules, enabling the generation of more compact rule representations for the decision boundaries of the classification problem.

Global and local explanations. Additionally, data analysts can easily obtain local explanations of specific instances, in the form of the fired decision rules. In addition, analysts can also obtain a global view of the relationships between the most important feature interactions using a complete ruleset. The global explainability of the simplified rules is complemented by the coverage and rule confidence of each rule.

7.4. Deep learning models

Even though the proposed method was developed to be applied to tree ensembles, which are considered traditional machine learning methods, in theory the algorithm can be applied to a set of neural networks, given that methods for rule extraction from neural networks are used. So far, the methods presented in the literature can work with small networks [31], because they were introduced before the rapid development of deep neural networks. Regardless, the rules extracted from neural networks can be combined and simplified using RuleCOSI+ algorithm. This topic is out of the scope of this paper, and it is left as a possible extension of our work.

7.5. Differences with the previous version

The original RuleCOSI algorithm [9] works with a specific set of problems. First, the accepted input of the algorithm was limited to only tree ensembles built with the AdaBoost algorithm and some variations, such as SMOTEBoost and RUSBoost. Second, the machine learning task performed by the simplified rule-based model was binary classification. Third, the algorithm was specialized to handle imbalanced data only. In contrast, RuleCOSI+ is presented as a general classification algorithm that can work with balanced data and other types of tree ensembles.

From an algorithmic point of view, RuleCOSI also has three main phases: extraction, combination, and simplification. However, RuleCOSI+ differs in the first two phases. The new extraction process is more complex because it must consider different tree ensemble frameworks. In addition, the proposed method can handle class probability estimates, whereas RuleCOSI only works with crisp rules. The combination process is completely renewed. RuleCOSI needed a combination matrix and a weigh vector to guide the combination process and choose the predicted class for each new combined rule. In contrast, RuleCOSI+ directly combines the class probability estimates, which makes the method faster and more accurate. In addition, the proposed method incorporates a sequential covering heuristic that was not present in the original algorithm. The simplification phase is very similar in both versions.

We compared the results presented in [9] using the enhanced algorithm over 20 imbalanced datasets (we excluded three private datasets). Because the best results with RuleCOSI were achieved by the simplified rules obtained from the RUSBoost variant, we used those results for our comparison. We ran RuleCOSI+ using the GBM algorithm with the same number of trees and an exponential loss function to recover the AdaBoost algorithm [32]. The summary of tie-win-loss scores using the Wilcoxon signed-rank test is presented in Figure 8. In the figure the algorithm is denoted as ADB, from AdaBoost. As observed from the results, ADB-RC+ is superior in all the measures to its predecessor. Moreover, the execution time reduction between both experiments is clearly significant: RuleCOSI averaged 6,464 s among all the experiments, whereas RuleCOSI+ performed the same experiments in 0.08 s on average.

7.6. Execution times

The execution times are divided into training (or simplification time) and inference time. The training time applies for learning algorithms such as ensemble models, DT, SKR and BSTR, while the simplification time applies for post-hoc simplification models. The summary of execution times is presented in Figure 9. The proposed method ranked last in this category since it took more time for tree ensemble simplification. This is a limitation of RuleCOSI+. However, this time can be reduced by controlling the coverage threshold parameter β , because this parameter helps to filter rules with low coverage, thus speeding up the

simplification process.

Finally, one of the benefits of generating small rulesets is the inference time. The inference time of the experiments is presented in the lower part of Figure 9. Prediction times are important for real-time systems that need to make predictions as fast as possible. RuleCOSI+ had shorter inference time compared with that of other similar methods thus increasing the applicability of the method to real-world problems.

8. Conclusion and future work

In this paper, we presented a method for combining and simplifying tree ensembles into a single interpretable set of decision rules. The algorithm works with both independent and dependent tree ensembles and increases the interpretability of the models as measured by two proposed interpretability measures, called REDU and UNIQ. The first measure helps to quantify the percentage of the size reduction between the two models, and the second one estimates the percentage of non-redundant information in the model.

Several experiments were performed to evaluate and validate the results, using six tree ensemble algorithms, four benchmark ensemble simplification methods and three single interpretable baseline models. The results are satisfactory, and further improve the study of the construction of accurate and interpretable models.

This study has some limitations, such as a lack of support for multiple classes. Additionally, the sizes of the datasets were limited to those used the experiments. The proposed method can be extended in different ways. One interesting direction is to add the regression task to the method; this is not trivial, because in the combination process, the mechanisms to calculate the predicted value must be investigated. Finally, it would be worth exploring different heuristics for rule evaluation and pruning, such as the J-measure or m-estimate, so as to evaluate the corresponding performance behaviors of the algorithm.

CRedit author statement

Josue Obregon: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – Original draft preparation, Visualization. **Jae-Yoon Jung:** Conceptualization, Methodology, Validation, Writing – Review & Editing, Visualization, Supervision, Project administration, Funding acquisition (Appendix A, Appendix B, Eqn 1-Eqn 5)

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Open datasets

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Appendix A. Other classification performance of all algorithms

Appendix para (Table A1, Table A2)

Table A1

Accuracy, precision and recall values of the used algorithms for RF and GBM (continued).

Dataset	RF-RuleCOSI+			RF-inTrees			RF-DefragTrees			RF-RuleFit		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
australian	84.54 ± 4.05	81.08 ± 7.86	87.00 ±	81.82 ± 9.65	78.22 ±	69.74 ±	84.78 ± 3.18	82.82 ± 5.09	83.73 ±	85.51 ± 4.58	86.95 ±	79.72 ±
	↓	↓	8.39 ↑	↓	27.28	25.21 ↓	↓	↓	7.39	↓	5.91	8.24 ↓
bands	65.77 ± 6.74	68.02 ±	89.71 ±	66.31 ± 8.42	72.10 ±	77.59 ±	63.47 ± 7.93	66.80 ± 6.72	85.22 ±	64.30 ± 3.38	64.92 ± 3.73	95.94 ±
	↓	6.97	13.87	↓	8.79	15.33 ↓	↓	↓	10.38 ↓	↓	↓	7.03 ↑
banknote	98.86 ±	98.40 ±	99.07 ±	98.74 ±	98.50 ±	98.69 ±	96.21 ± 1.54	95.06 ± 2.65	96.61 ±	99.54 ±	99.30 ±	99.67 ±
	0.83	1.44	1.34	1.03	1.67	1.80 ↓	↓	↓	3.39 ↓	0.59 ↑	1.08 ↑	0.67
biodeg	84.14 ± 4.99	86.74 ±	90.61 ±	84.41 ±	89.58 ±	86.82 ±	77.70 ± 4.35	81.63 ± 4.14	85.98 ±	84.87 ± 3.11	89.94 ±	86.94 ±
	↓	5.71	5.04	3.62	4.18 ↑	4.20 ↓	↓	↓	5.90 ↓	↓	2.63 ↑	3.55 ↓
cardiotcg	94.28 ±	95.30 ±	97.48 ±	94.07 ±	94.61 ±	97.96 ±	90.56 ± 1.74	92.52 ± 1.91	95.67 ±	94.06 ±	96.51 ±	95.85 ±
	1.78	1.49 ↑	1.69 ↓	1.67	1.43	1.33	↓	↓	1.78 ↓	1.55	1.25 ↑	1.74 ↓
credit	99.80 ±	97.86 ±	100.00 ±	99.70 ±	96.30 ±	98.33 ±	99.60 ±	97.38 ±	98.00 ±	99.75 ±	97.75 ±	99.33 ±
	0.52 ↑	5.57	0.00 ↑	0.74 ↑	10.29	9.13 ↑	0.68	5.98	6.10 ↑	0.69 ↑	5.89	3.65 ↑
cryotherapy	87.41 ±	89.82 ±	87.50 ±	87.22 ±	86.72 ±	90.83 ±	87.41 ± 9.10	93.16 ±	83.67 ±	91.48 ±	93.67 ±	90.00 ±
	11.57 ↓	12.72 ↓	15.74	10.90	16.28 ↓	15.44	↓	10.36	16.40	9.97	10.05	14.26
divorce	96.86 ±	98.43 ±	95.23 ±	96.27 ±	98.02 ±	94.85 ±	97.45 ±	99.21 ±	95.56 ±	97.45 ±	99.63 ±	95.19 ±
	4.01	4.09	6.63	4.50	5.37	8.14	3.34	3.00	5.95	2.96	2.03	6.01
ecoli0vs1	98.48 ±	98.26 ±	99.52 ±	97.73 ±	97.64 ±	99.10 ±	98.79 ±	98.26 ±	100.00 ±	98.33 ±	98.69 ±	98.86 ±
	2.18	2.93	1.81	3.73	3.92	3.83	2.04	2.93	0.00	2.53	2.66	3.14
ecoli1	88.60 ±	92.50 ±	93.17 ±	88.94 ±	92.16 ±	93.78 ±	89.59 ±	92.39 ±	94.73 ±	88.98 ±	93.70 ±	92.17 ±
	5.27	5.15	6.69	4.33	3.65	5.10	4.93	5.31	5.10 ↑	5.55	4.50	5.66
ecoli2	93.86 ±	96.00 ±	96.83 ±	93.23 ± 3.48	95.07 ±	97.20 ±	93.57 ±	95.48 ±	97.09 ±	94.55 ±	96.03 ±	97.66 ±
	2.99	2.43	3.09 ↓	↓	3.25	3.46 ↓	3.30	3.00	3.02 ↓	3.14	2.61	2.51 ↓
ecoli3	93.05 ±	95.00 ±	97.46 ±	91.73 ±	94.09 ±	96.91 ±	92.47 ±	94.87 ±	96.90 ±	93.05 ±	94.54 ±	98.01 ±
	3.15	2.51	2.71 ↓	4.77	4.23	3.82	3.27	2.46	3.14 ↓	2.97	2.71	2.55
glass	93.76 ±	95.78 ±	96.30 ±	93.38 ±	96.32 ±	95.23 ±	92.53 ±	94.78 ±	95.71 ±	93.61 ±	96.64 ±	95.10 ±
	5.27	4.59	5.81	5.94	5.34	5.18	6.85	5.42	5.82	5.00	3.93	5.71
glass0	86.12 ±	89.83 ±	90.03 ±	82.83 ±	87.26 ± 7.89	88.05 ±	78.41 ±	86.83 ± 9.40	80.83 ±	83.63 ± 9.07	89.65 ±	85.86 ±
	6.45	6.31	6.80	7.66	↓	9.20	11.53 ↓	↓	13.45 ↓	↓	6.62	10.88 ↓
glass1	82.09 ±	86.29 ±	86.63 ±	79.62 ±	82.12 ±	88.51 ±	72.40 ± 9.65	77.96 ± 9.42	81.87 ±	78.00 ±	84.77 ±	80.86 ±
	9.10	8.24 ↑	12.39 ↓	8.86	9.01	7.74 ↓	↓	↓	12.66 ↓	11.45 ↓	9.83	12.27 ↓
glass6	95.66 ±	96.94 ±	98.22 ±	96.26 ±	96.65 ±	99.29 ±	95.50 ±	97.28 ±	97.68 ±	96.41 ±	97.59 ±	98.37 ±
	3.83	3.39	2.57	3.73	3.72	1.85	3.78	3.45	3.03 ↓	4.58	3.53	2.53
haberman	73.63 ±	74.52 ±	97.92 ±	72.62 ±	73.48 ±	98.37 ±	75.40 ±	78.78 ±	91.77 ±	74.18 ±	75.29 ±	97.67 ±
	2.13	1.75	4.69	8.41	8.38	2.90	6.27 ↑	4.81 ↑	5.11 ↓	2.35	4.03	5.46
heart	80.12 ± 7.56	78.96 ± 8.11	88.89 ±	72.26 ±	71.17 ±	87.18 ±	79.26 ± 8.17	80.53 ± 7.85	83.33 ±	79.75 ± 8.58	84.93 ±	77.56 ±
	↓	↓	7.89	10.20 ↓	11.77 ↓	13.50	↓	↓	10.76 ↓	↓	8.25	10.86 ↓
ionosphere	91.46 ± 4.59	91.76 ±	95.71 ±	90.20 ± 5.49	90.40 ±	94.91 ±	90.71 ± 5.29	91.79 ±	94.09 ±	91.45 ± 4.69	93.03 ±	94.05 ±
	↓	5.78	3.56 ↓	↓	6.39	5.43 ↓	↓	4.53	5.20 ↓	↓	5.05	5.72 ↓
maintenance	97.94 ±	81.68 ±	53.60 ±	97.84 ±	80.78 ±	48.49 ±	97.18 ± 0.28	79.61 ±	24.41 ±	97.90 ±	80.56 ± 8.44	51.07 ±
	0.36 ↑	11.11 ↓	13.17 ↑	0.53 ↑	10.12 ↓	12.10 ↑	↓	14.36 ↓	9.89 ↓	0.35 ↑	↓	10.09 ↑
mammographic	82.65 ± 4.17	81.24 ± 5.95	84.37 ±	52.91 ± 5.57	31.88 ±	30.82 ±	82.93 ±	83.21 ±	81.88 ±	84.18 ±	87.89 ±	78.23 ±
	↓	↓	7.34	↓	35.37 ↓	45.30 ↓	4.42	6.25	6.76	4.42	4.84 ↑	6.60 ↓
newthyroid1	97.19 ±	98.42 ±	98.33 ±	97.97 ±	98.16 ±	99.39 ±	97.69 ±	98.28 ±	99.07 ±	98.13 ±	98.42 ±	99.44 ±
	3.61	2.79	3.31	2.93	3.13	1.89	3.37	3.09	2.95	2.33	2.45	1.70
newthyroid2	97.07 ±	97.80 ±	98.89 ±	96.73 ±	97.33 ±	98.90 ±	96.90 ±	98.08 ±	98.33 ±	97.68 ±	97.92 ±	99.44 ±
	3.70	3.83	2.69	4.12	3.29	3.01	3.04	2.89	2.97 ↓	3.38	3.19	2.24
occupancy	99.34 ±	99.61 ± 0.24	99.55 ±	99.27 ±	99.70 ±	99.37 ±	98.76 ± 0.35	99.62 ± 0.41	98.81 ±	99.34 ±	99.73 ±	99.43 ±
	0.26	↓	0.26 ↑	0.24	0.19	0.36	↓	↓	0.40 ↓	0.20	0.22	0.27 ↑
pageblocks	97.31 ±	98.68 ±	98.33 ±	96.78 ± 0.69	97.54 ±	98.91 ±	95.67 ± 0.90	97.16 ± 0.96	98.06 ±	97.06 ±	98.45 ±	98.28 ±
	0.63	0.44 ↑	0.69 ↓	↓	0.69	0.64 ↓	↓	↓	0.77 ↓	0.79	0.57 ↑	0.67 ↓
pima	73.17 ± 4.20	75.15 ±	88.93 ±	68.49 ± 6.55	71.17 ± 8.40	89.56 ±	73.48 ± 4.29	76.69 ±	85.73 ±	75.30 ±	83.53 ±	77.47 ±
	↓	4.37	7.73	↓	↓	10.99	↓	4.39	6.47 ↓	4.29	3.90 ↑	4.55 ↓
segment0	99.51 ±	99.65 ±	99.78 ±	99.57 ±	99.75 ±	99.75 ±	99.09 ± 0.58	99.51 ± 0.45	99.43 ±	99.62 ±	99.75 ±	99.81 ±
	0.41	0.40	0.37	0.48	0.39	0.37 ↓	↓	↓	0.54 ↓	0.41	0.32	0.34

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Table A1 (continued)

Dataset	RF-RuleCOSI+			RF-inTrees			RF-DefragTrees			RF-RuleFit		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
tic-tac-toe	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
vehicle2	97.36 ± 2.52	98.47 ± 1.47	97.98 ± 2.73	97.00 ± 2.17	98.39 ± 1.63	97.54 ± 2.44	93.62 ± 3.17	95.58 ± 2.97	95.97 ± 3.76	97.91 ± 1.54	98.63 ± 0.99	98.57 ± 1.69
vehicle3	79.24 ± 3.19	80.65 ± 4.14	95.80 ± 5.24	78.39 ± 3.75	81.43 ± 4.73	92.60 ± 5.54	76.79 ± 3.26	78.61 ± 2.07	195.01 ± 5.34	75.81 ± 2.26	79.84 ± 5.60	92.27 ± 8.68
wisconsin	95.46 ± 2.53	96.95 ± 2.24	96.09 ± 3.34	94.82 ± 3.40	95.34 ± 5.13	96.90 ± 2.75	95.65 ± 2.91	97.30 ± 2.36	96.01 ± 3.35	96.68 ± 2.33	98.41 ± 1.80	96.47 ± 2.88
yeast1	75.47 ± 3.09	77.22 ± 2.94	93.21 ± 3.85	72.56 ± 3.56	74.20 ± 4.10	94.34 ± 3.91	74.91 ± 2.87	79.21 ± 2.94	88.06 ± 4.84	73.70 ± 2.99	78.17 ± 6.01	89.44 ± 8.82
yeast3	94.83 ± 1.73	97.40 ± 1.35	96.80 ± 1.36	92.60 ± 2.39	94.97 ± 2.95	96.98 ± 2.41	94.86 ± 1.53	97.14 ± 1.30	97.10 ± 1.52	95.22 ± 1.54	97.70 ± 1.21	96.92 ± 1.54
Average	90.15 ± 3.56	90.74 ± 4.19	93.60 ± 4.93	88.25 ± 4.35	88.22 ± 6.76	90.81 ± 7.10	88.89 ± 3.88	90.23 ± 4.33	90.80 ± 5.28	90.23 ± 3.44	91.89 ± 3.81	91.99 ± 4.92

Table A2
Accuracy, precision and recall values of the used algorithms for RF and GBM (continued).

Dataset	RF-FBT			SKR			DT			GBM-RuleCOSI+		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
australian	84.83 ± 3.61	82.65 ± 6.54	84.91 ± 10.57	73.77 ± 9.56	80.66 ± 32.74	45.27 ± 23.93	85.22 ± 2.86	78.98 ± 4.65	91.76 ± 6.04	84.73 ± 3.16	80.66 ± 5.48	87.31 ± 6.66
bands	63.02 ± 1.73	63.94 ± 2.07	95.80 ± 9.17	40.08 ± 5.20	38.29 ± 46.12	5.80 ± 9.30	58.66 ± 5.58	63.88 ± 3.82	79.86 ± 12.43	64.84 ± 5.76	67.51 ± 5.01	87.39 ± 11.27
banknote	97.81 ± 2.77	97.59 ± 2.84	97.60 ± 5.72	81.48 ± 10.57	92.64 ± 17.90	61.75 ± 25.82	98.08 ± 1.33	97.68 ± 1.77	98.03 ± 1.85	98.71 ± 1.00	98.39 ± 1.52	98.74 ± 1.53
biodeg	77.61 ± 6.27	81.11 ± 7.83	87.89 ± 7.83	41.01 ± 13.03	37.82 ± 42.90	12.48 ± 21.20	81.99 ± 3.61	86.14 ± 3.40	86.98 ± 4.48	85.15 ± 4.19	86.62 ± 4.13	92.04 ± 3.89
cardiotog	85.70 ± 3.47	89.47 ± 4.13	93.03 ± 6.14	83.06 ± 12.20	95.36 ± 1.74	82.43 ± 16.53	93.35 ± 1.86	94.81 ± 1.76	96.80 ± 1.61	94.93 ± 1.25	95.71 ± 1.33	97.90 ± 1.16
credit	99.85 ± 0.46	98.97 ± 3.94	99.44 ± 3.04	99.55 ± 1.20	98.97 ± 3.94	95.44 ± 15.37	99.85 ± 0.46	98.41 ± 4.86	100.00 ± 0.00	99.90 ± 0.38	98.97 ± 3.94	100.00 ± 0.00
cryotherapy	88.52 ± 7.43	94.82 ± 9.84	84.50 ± 13.28	79.26 ± 13.91	89.33 ± 30.51	61.00 ± 28.96	90.00 ± 11.05	91.94 ± 12.31	89.50 ± 14.10	89.26 ± 9.89	90.99 ± 11.99	89.50 ± 14.10
divorce	97.84 ± 2.88	98.93 ± 3.28	96.71 ± 5.55	97.25 ± 3.70	100.00 ± 0.00	94.40 ± 7.52	98.04 ± 3.22	98.88 ± 3.43	97.13 ± 5.30	97.45 ± 3.34	98.65 ± 4.25	96.44 ± 5.55
ecoli0vs1	98.48 ± 2.18	98.04 ± 3.04	99.78 ± 1.22	96.52 ± 6.17	98.62 ± 2.82	95.97 ± 9.12	98.64 ± 2.12	98.04 ± 3.04	100.00 ± 0.00	98.64 ± 2.12	98.04 ± 3.04	100.00 ± 0.00
ecoli1	89.80 ± 4.95	91.79 ± 5.78	95.90 ± 4.62	81.44 ± 16.55	95.49 ± 18.14	76.99 ± 21.64	90.07 ± 5.38	91.74 ± 4.87	96.14 ± 5.62	89.69 ± 4.95	91.27 ± 4.49	96.14 ± 5.45
ecoli2	93.97 ± 3.91	94.96 ± 3.69	98.24 ± 2.21	26.67 ± 19.17	35.60 ± 47.68	13.73 ± 23.83	93.56 ± 3.64	94.97 ± 3.21	97.66 ± 2.79	93.16 ± 3.10	95.07 ± 3.06	97.07 ± 3.20

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Table A2 (continued)

Dataset	RF-FBT			SKR			DT			GBM-RuleCOSI+		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
ecoli3	90.59 ± 2.66 ↓	91.08 ± 2.92 ↓	99.34 ± 1.61 ↑	63.73 ± 26.35 ↓	85.82 ± 34.28 ↑	60.15 ± 29.84 ↓	91.57 ± 3.13 ↓	93.55 ± 3.03 ↓	97.46 ± 3.46	92.06 ± 3.06	93.44 ± 2.55	98.12 ± 2.56
glass	93.29 ± 5.47	96.40 ± 4.80	95.07 ± 6.19	92.53 ± 5.53	98.95 ± 2.90 ↑	91.24 ± 6.71 ↓	93.90 ± 4.82	95.95 ± 4.39	96.31 ± 5.29	93.76 ± 4.47	96.11 ± 4.05	95.91 ± 5.47
glass0	81.18 ± 8.19 ↓	88.54 ± 8.16 ↓	83.79 ± 9.50 ↓	40.19 ± 10.46 ↓	42.67 ± 49.75 ↓	11.32 ± 15.73 ↓	80.90 ± 10.08 ↓	87.64 ± 8.56 ↓	84.10 ± 9.81 ↓	82.21 ± 8.91	86.58 ± 7.49	87.89 ± 10.73
glass1	75.25 ± 9.33 ↓	77.29 ± 9.41 ↓	89.62 ± 9.48 ↓	60.47 ± 16.56 ↓	72.82 ± 31.20	50.79 ± 32.07 ↓	73.98 ± 10.42 ↓	77.65 ± 7.21 ↓	83.97 ± 13.87 ↓	78.82 ± 10.27	82.94 ± 9.60	85.95 ± 11.73
glass6	95.33 ± 4.43	96.01 ± 4.30	98.93 ± 2.18	93.91 ± 5.57 ↓	98.39 ± 2.87 ↓	94.57 ± 5.56 ↓	93.61 ± 5.30 ↓	96.00 ± 3.78 ↓	96.78 ± 4.14 ↓	95.17 ± 3.97	96.24 ± 3.50	98.39 ± 2.50
haberman	72.95 ± 4.57	75.09 ± 3.49	95.26 ± 6.63	28.66 ± 5.03 ↓	34.44 ± 47.53 ↓	3.84 ± 6.43 ↓	74.94 ± 7.91	80.51 ± 5.87 ↑	87.62 ± 5.64 ↓	73.09 ± 3.98	74.51 ± 2.32 ↓	96.75 ± 5.18 ↑
heart	75.31 ± 7.30 ↓	72.83 ± 7.02 ↓	90.00 ± 10.02	53.46 ± 13.00 ↓	40.01 ± 44.65 ↓	19.78 ± 26.17 ↓	78.15 ± 9.60 ↓	78.62 ± 9.82 ↓	84.67 ± 11.37 ↓	79.51 ± 7.26	79.15 ± 8.89 ↓	88.00 ± 9.65
ionosphere	86.42 ± 5.11 ↓	86.37 ± 5.99 ↓	94.50 ± 7.37 ↓	88.51 ± 10.04 ↓	93.17 ± 4.56 ↓	89.08 ± 17.16 ↓	89.27 ± 3.93 ↓	92.09 ± 4.99	91.52 ± 6.73 ↓	89.46 ± 4.98 ↓	90.18 ± 6.28 ↓	94.51 ± 5.05 ↓
maintenance	97.16 ± 0.32 ↓	78.85 ± 30.78	19.03 ± 10.57 ↓	96.69 ± 0.11 ↓	39.35 ± 44.37 ↓	3.24 ± 4.26 ↓	97.92 ± 0.42 ↑	69.30 ± 7.01 ↓	70.31 ± 7.38 ↑	98.53 ± 0.34	86.16 ± 7.40	68.36 ± 8.54
mammographic	83.65 ± 4.02	82.14 ± 5.12	85.19 ± 6.03 ↑	68.15 ± 14.52 ↓	63.35 ± 42.63	38.85 ± 33.37 ↓	83.78 ± 4.13	87.11 ± 6.12 ↑	78.73 ± 7.52 ↓	82.05 ± 4.20 ↓	80.59 ± 5.52 ↓	83.53 ± 5.77 ↑
newthyroid1	91.65 ± 5.84 ↓	91.73 ± 5.93 ↓	99.44 ± 2.24	89.94 ± 15.16 ↓	96.11 ± 18.23	88.52 ± 18.57 ↓	96.12 ± 3.48 ↓	97.87 ± 2.66 ↓	97.59 ± 3.77 ↓	97.06 ± 2.57	97.54 ± 2.67	99.07 ± 2.56
newthyroid2	95.99 ± 3.56 ↓	96.04 ± 3.71 ↓	99.44 ± 1.70	90.11 ± 6.38 ↓	98.74 ± 2.32 ↓	89.44 ± 8.67 ↓	95.98 ± 3.79 ↓	97.92 ± 3.38	97.41 ± 3.79 ↓	96.77 ± 3.45	97.60 ± 3.64	98.70 ± 2.39
occupancy	98.74 ± 0.36 ↓	99.74 ± 0.27	98.66 ± 0.51 ↓	98.76 ± 0.33 ↓	99.86 ± 0.13 ↓	98.57 ± 0.45 ↓	99.32 ± 0.25 ↓	99.64 ± 0.23 ↓	99.49 ± 0.27 ↑	99.29 ± 0.27	99.53 ± 0.36 ↓	99.57 ± 0.23
pageblocks	93.52 ± 2.47 ↓	94.06 ± 2.92 ↓	99.17 ± 0.75	94.69 ± 1.52 ↓	98.06 ± 0.84 ↓	96.00 ± 1.82 ↓	97.08 ± 0.77	98.16 ± 0.67 ↑	98.60 ± 0.59 ↓	97.26 ± 0.65	98.08 ± 0.48 ↓	98.88 ± 0.53 ↑
pima	73.95 ± 3.77 ↓	75.43 ± 3.66 ↓	89.53 ± 5.40	48.86 ± 8.41 ↓	81.15 ± 33.73	22.87 ± 13.81 ↓	73.26 ± 5.31 ↓	76.70 ± 4.64	85.20 ± 6.59 ↓	74.95 ± 5.66	76.03 ± 4.53	90.40 ± 5.05 ↑
segment0	88.19 ± 4.36 ↓	88.43 ± 4.93 ↓	99.61 ± 1.28	77.93 ± 12.75 ↓	96.24 ± 18.18	74.58 ± 14.89 ↓	99.21 ± 0.59 ↓	99.46 ± 0.42 ↓	99.61 ± 0.51 ↓	99.42 ± 0.54 ↓	99.61 ± 0.43 ↓	99.71 ± 0.43
tic-tac-toe	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.58 ± 2.28	99.80 ± 1.07	99.21 ± 4.35	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
vehicle2	84.83 ± 7.20 ↓	88.37 ± 8.48 ↓	93.43 ± 9.05 ↓	38.90 ± 15.02 ↓	86.05 ± 34.38	17.96 ± 20.69 ↓	95.74 ± 2.35 ↓	97.42 ± 1.94 ↓	96.87 ± 2.23 ↓	97.05 ± 2.21 ↓	98.06 ± 1.72 ↓	97.99 ± 2.28 ↓
vehicle3	75.49 ± 1.65 ↓	75.66 ± 1.76 ↓	99.37 ± 2.45 ↑	31.20 ± 7.83 ↓	72.08 ± 42.80	8.98 ± 11.57 ↓	76.95 ± 2.27 ↓	79.79 ± 4.12	93.58 ± 7.08	78.17 ± 2.89	79.67 ± 3.46 ↓	95.64 ± 4.24 ↑
wisconsin	92.54 ± 3.48 ↓	93.38 ± 3.45 ↓	95.42 ± 3.76 ↓	94.58 ± 3.38 ↓	98.57 ± 2.22 ↓	93.08 ± 5.26 ↓	95.17 ± 2.65 ↓	96.53 ± 2.67 ↓	96.09 ± 2.71 ↓	95.70 ± 2.42 ↓	96.82 ± 2.30	96.61 ± 2.71
yeast1	74.08 ± 2.83 ↓	77.33 ± 2.94	90.36 ± 5.63 ↓	34.09 ± 5.58 ↓	83.92 ± 34.09 ↑	7.70 ± 8.18 ↓	74.89 ± 3.31 ↓	79.71 ± 3.44 ↑	87.17 ± 5.10 ↓	76.75 ± 2.68 ↓	78.05 ± 1.92 ↓	93.71 ± 2.93 ↑
yeast3	94.81 ± 1.68	96.31 ± 1.48	97.96 ± 1.45	40.67 ± 30.89 ↓	76.63 ± 43.00	33.37 ± 34.71 ↓	94.86 ± 1.98	97.50 ± 1.33 ↑	96.72 ± 1.96 ↓	95.01 ± 1.96	97.17 ± 1.39 ↑	97.25 ± 1.90 ↓
Average	87.65 ± 3.89	88.28 ± 5.26	92.33 ± 5.25	70.48 ± 9.94	79.36 ± 23.64	55.71 ± 15.86	89.21 ± 3.87	90.14 ± 4.04	92.54 ± 4.97	89.96 ± 3.51	90.48 ± 3.90	94.17 ± 4.40

Table A3

Accuracy, precision and recall values of the used algorithms for RF and GBM.

Dataset	GBM-inTrees			GBM-DefragTrees			GBM-RuleFit			GBM-FBT			GBM-Boosted Rules		
	Accuracy	Precision	Recall	Accuracy	Accuracy	Precision	Recall	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
australian	61.98 ±	54.13 ±	19.46 ±	84.93 ±	85.41 ±	78.76 ±	92.41 ±	82.89 ±	84.16 ±	86.67 ±	88.80 ±	80.47 ±	83.24 ±	76.45 ±	93.61 ±
	9.36 ↓	41.80 ↓	21.45 ↓	3.66	2.61	3.54 ↓	4.37 ↑	5.74	8.51	3.28	4.90 ↑	6.02 ↓	8.78	8.47 ↓	4.38 ↑
bands	67.48 ±	66.71 ±	97.13 ±	62.56 ±	65.75 ±	70.43 ±	81.30 ±	66.44 ±	83.33 ±	64.20 ±	64.57 ±	96.52 ±	60.64 ±	65.08 ±	82.61 ±
	6.71	6.44 ↓	4.65 ↑	5.23 ↓	6.24 ↓	6.16	16.27	4.05 ↓	10.84	3.52 ↓	2.68 ↓	5.98 ↑	5.95 ↓	4.70 ↓	17.05
banknote	96.35 ±	93.86 ±	98.47 ±	96.33 ±	92.37 ±	99.09 ±	83.66 ±	95.47 ±	96.45 ±	99.71 ±	99.57 ±	99.78 ±	99.03 ±	98.76 ±	99.07 ±
	2.32 ↓	4.29 ↓	1.93 ↓	1.86 ↓	3.87 ↓	2.37	8.50 ↓	3.26 ↓	2.55 ↓	0.49	1.01	0.57	1.00 ↓	1.44 ↓	1.41 ↓
biodeg	67.05 ±	67.35 ±	98.00 ±	79.12 ±	75.26 ±	73.74 ±	97.66 ±	83.27 ±	86.22 ±	85.18 ±	90.15 ±	87.27 ±	76.53 ±	82.95 ±	83.65 ±
	3.05 ↓	2.73 ↓	2.51 ↑	4.28 ↓	2.64 ↓	2.64 ↓	2.82 ↑	4.59 ↓	5.54 ↓	3.22 ↓	2.82 ↑	4.31 ↓	6.98 ↓	7.88 ↓	11.54 ↓
cardiotcg	88.86 ±	89.11 ±	97.81 ±	90.03 ±	91.05 ±	94.65 ±	93.89 ±	92.71 ±	94.70 ±	94.45 ±	96.42 ±	96.45 ±	81.66 ±	83.05 ±	97.10 ±
	3.45 ↓	3.60 ↓	1.11	2.00 ↓	2.23 ↓	2.28 ↓	3.16 ↓	2.02 ↓	2.71 ↓	1.73 ↓	1.08	1.87 ↓	4.45 ↓	5.88 ↓	4.48
credit	99.85 ±	98.08 ±	100.00 ±	99.90 ±	99.70 ±	96.83 ±	100.00 ±	98.97 ±	100.00 ±	99.85 ±	98.41 ±	100.00 ±	99.85 ±	98.41 ±	100.00 ±
	0.46	6.71	0.00	0.38	0.61	6.48	0.00	3.94	0.00	0.46	4.86	0.00	0.46	4.86	0.00
cryotherapy	88.84 ±	92.18 ±	86.42 ±	88.89 ±	90.00 ±	95.17 ±	86.67 ±	94.64 ±	85.83 ±	91.11 ±	95.61 ±	87.50 ±	88.15 ±	91.13 ±	89.83 ±
	10.52	11.79	17.82	9.68	8.92	9.37	13.85	10.25	16.97	9.85	9.05 ↑	14.84	14.86	15.32	13.80
divorce	95.69 ±	96.75 ±	94.85 ±	96.86 ±	96.67 ±	98.02 ±	95.56 ±	97.47 ±	96.39 ±	97.65 ±	100.00 ±	95.19 ±	98.24 ±	99.63 ±	96.76 ±
	4.87 ↓	6.44	6.85	3.70	3.68	5.37	5.95	4.68	5.62	2.93	0.00	6.01	3.15	2.03	5.48
ecoli0vs1	98.79 ±	98.29 ±	100.00 ±	98.79 ±	98.64 ±	98.04 ±	100.00 ±	98.47 ±	99.76 ±	98.18 ±	98.23 ±	99.08 ±	98.64 ±	98.04 ±	100.00 ±
	2.37	3.32	0.00	2.04	2.12	3.04	0.00	2.82	1.30	2.82	2.98	2.96	2.12	3.04	0.00
ecoli1	81.30 ±	81.63 ±	98.73 ±	89.39 ±	80.75 ±	80.40 ±	99.62 ±	93.83 ±	92.68 ±	89.50 ±	94.19 ±	92.29 ±	89.58 ±	95.05 ±	91.66 ±
	5.61 ↓	6.36 ↓	2.87 ↑	5.29	3.90 ↓	3.75 ↓	1.55 ↑	5.04 ↑	5.85 ↓	4.93	4.01 ↑	5.88 ↓	5.31	4.59 ↑	7.63 ↓
ecoli2	86.68 ±	88.40 ±	97.67 ±	92.76 ±	93.17 ±	93.05 ±	99.53 ±	94.73 ±	96.96 ±	94.26 ±	96.45 ±	96.84 ±	91.77 ±	95.62 ±	94.72 ±
	3.88 ↓	5.56 ↓	4.31	3.55 ↓	4.03	3.97 ↓	1.21 ↑	3.15	3.77	2.89	2.58	2.65 ↓	5.46 ↓	3.45	5.31 ↓
ecoli3	91.06 ±	91.66 ±	98.95 ±	92.57 ±	92.86 ±	93.52 ±	99.01 ±	94.43 ±	97.57 ±	93.26 ±	94.62 ±	98.13 ±	86.04 ±	95.10 ±	89.47 ±
	5.34	4.40 ↓	3.59	3.52	2.76	2.88	1.97	2.79	3.69	3.21	2.51	2.71	5.82 ↓	4.20	9.33 ↓
glass	90.56 ±	95.73 ±	92.22 ±	93.30 ±	90.17 ±	90.96 ±	97.12 ±	95.89 ±	95.51 ±	94.25 ±	96.92 ±	95.72 ±	85.00 ±	91.75 ±	83.91 ±
	5.96	5.14	8.45	5.01	5.69 ↓	5.51 ↓	4.50	4.45	5.31	4.97	4.13	5.36	20.19 ↓	19.80	27.52 ↓
glass0	75.25 ±	84.10 ±	80.22 ±	81.17 ±	73.99 ±	74.94 ±	93.54 ±	87.68 ±	84.52 ±	83.95 ±	89.81 ±	86.35 ±	78.69 ±	87.81 ±	80.10 ±
	9.78 ↓	9.68	14.07 ↓	9.97 ↓	7.35 ↓	6.60 ↓	7.55 ↑	8.44	10.92 ↓	9.65	7.84	10.17 ↓	10.79 ↓	8.88	14.35 ↓
glass1	69.79 ±	73.64 ±	85.56 ±	73.64 ±	74.63 ±	75.32 ±	92.25 ±	78.99 ±	83.24 ±	80.69 ±	85.47 ±	85.07 ±	73.32 ±	76.80 ±	86.61 ±
	9.67 ↓	9.40 ↓	12.35	9.26 ↓	8.39 ↓	8.58 ↓	8.29 ↑	9.90 ↓	12.42 ↓	10.59	8.96	12.14 ↓	9.85 ↓	9.24 ↓	12.36
glass6	94.47 ±	95.03 ±	99.09 ±	95.81 ±	96.74 ±	97.64 ±	98.75 ±	97.25 ±	98.04 ±	95.63 ±	97.61 ±	97.48 ±	90.41 ±	97.07 ±	91.96 ±
	4.54	4.75	2.07	4.32	3.70	3.38	2.30	3.21	3.58	4.62	3.51	3.98	15.32 ↓	3.83	17.90 ↓
haberman	73.49 ±	73.65 ±	99.69 ±	74.61 ±	75.51 ±	76.67 ±	97.21 ±	78.42 ±	91.03 ±	74.73 ±	75.58 ±	97.92 ±	71.55 ±	80.78 ±	81.26 ±
	8.37	8.27	1.18 ↑	6.22	6.25 ↑	6.09	4.80 ↑	4.79 ↑	5.93	4.45	4.46	4.47 ↑	9.18	5.89 ↑	12.45 ↓
heart	55.00 ±	55.49 ±	97.21 ±	78.27 ±	82.47 ±	84.99 ±	84.22 ±	78.47 ±	85.56 ±	80.00 ±	83.36 ±	80.44 ±	77.28 ±	77.53 ±	85.56 ±
	9.66 ↓	9.66 ↓	4.59 ↑	9.42 ↓	6.22	7.49 ↑	9.51	9.88 ↓	9.11	8.06	8.31	9.58 ↓	8.00 ↓	9.52 ↓	9.60
ionosphere	91.63 ±	92.27 ±	95.08 ±	90.98 ±	88.99 ±	91.16 ±	92.30 ±	91.28 ±	95.55 ±	92.69 ±	93.47 ±	95.53 ±	88.42 ±	88.87 ±	94.81 ±
	4.53	5.42	4.51	5.02	5.10 ↓	6.18	5.59 ↓	5.96	4.36	4.09	4.63	4.60	6.01 ↓	7.69 ↓	6.57
maintenance	98.14 ±	80.19 ±	60.52 ±	97.09 ±	97.00 ±	74.63 ±	22.33 ±	75.66 ±	21.63 ±	98.56 ±	86.70 ±	68.46 ±	96.69 ±	61.38 ±	9.43 ±
	0.39 ↓	7.96 ↓	11.89 ↓	0.33 ↓	0.37 ↓	19.17 ↓	7.71 ↓	16.07 ↓	9.92 ↓	0.37	7.22	7.93	0.56 ↓	40.54 ↓	9.05 ↓

(continued on next page)

Table A3 (continued)

Dataset	GBM-inTrees			GBM-DefragTrees			GBM-RuleFit			GBM-FBT			GBM-Boosted Rules		
	Accuracy	Precision	Recall	Accuracy	Accuracy	Precision	Recall	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
mammographic	50.78 ±	3.06 ±	6.44 ±	83.05 ±	82.81 ±	89.98 ±	72.86 ±	84.74 ±	79.72 ±	83.65 ±	87.67 ±	77.23 ±	81.89 ±	82.80 ±	79.90 ±
	5.67 ↓	12.03 ↓	24.54 ↓	3.78	4.00	5.53 ↑	5.78 ↓	4.97	6.68	4.41	4.97 ↑	6.54 ↓	5.06 ↓	6.60 ↓	6.28
newthyroid1	96.88 ±	97.63 ±	98.67 ±	98.76 ±	98.29 ±	98.42 ±	99.63 ±	99.12 ±	99.44 ±	98.29 ±	98.60 ±	99.44 ±	95.32 ±	97.08 ±	97.59 ±
	3.10	3.51	2.50	2.09	2.29	2.45	1.41	1.99 ↑	1.70	2.29	2.37	1.70	7.26	3.70	8.34
newthyroid2	96.75 ±	97.47 ±	98.69 ±	97.09 ±	98.00 ±	98.30 ±	99.44 ±	97.95 ±	98.70 ±	97.98 ±	97.93 ±	99.81 ±	95.38 ±	96.84 ±	97.96 ±
	3.88	3.05	2.82	3.06	3.11	3.27	1.70	3.42	2.39	2.89	3.18	1.01	4.35 ↓	4.82	3.09
occupancy	90.67 ±	93.69 ±	95.85 ±	98.92 ±	98.87 ±	99.82 ±	98.74 ±	99.71 ±	98.91 ±	99.42 ±	99.71 ±	99.55 ±	97.94 ±	99.86 ±	97.52 ±
	6.76 ↓	8.69	4.63 ↓	0.38 ↓	0.33 ↓	0.16 ↑	0.47 ↓	0.29	0.55 ↓	0.25	0.16 ↑	0.30	4.36 ↓	0.14 ↑	5.56 ↓
pageblocks	92.57 ±	92.79 ±	99.49 ±	95.67 ±	95.58 ±	97.02 ±	98.10 ±	97.19 ±	98.03 ±	97.48 ±	98.70 ±	98.49 ±	91.90 ±	92.13 ±	99.58 ±
	1.78 ↓	1.96 ↓	0.39 ↑	0.76 ↓	0.67 ↓	0.58 ↓	0.77 ↓	0.89 ↓	0.90 ↓	0.71	0.45 ↑	0.59	2.35 ↓	2.62 ↓	0.56 ↑
pima	65.24 ±	65.24 ±	99.73 ±	74.30 ±	73.26 ±	74.51 ±	89.73 ±	77.49 ±	85.53 ±	73.73 ±	82.52 ±	75.87 ±	72.09 ±	77.94 ±	81.53 ±
	5.52 ↓	5.54 ↓	0.86 ↑	3.57	3.81 ↓	2.74 ↓	4.39	2.76	5.65 ↓	4.55 ↓	3.81 ↑	5.56 ↓	6.26 ↓	5.80	14.57 ↓
segment0	99.19 ±	99.73 ±	99.32 ±	99.22 ±	98.66 ±	98.47 ±	100.00 ±	99.53 ±	99.56 ±	99.75 ±	99.83 ±	99.88 ±	91.84 ±	91.85 ±	99.66 ±
	0.54 ↓	0.39	0.59 ↓	0.60 ↓	0.95 ↓	1.06 ↓	0.00 ↑	0.44 ↓	0.62 ↓	0.35 ↑	0.30	0.25	4.92 ↓	5.02 ↓	0.48
tic-tac-toe	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±	100.00 ±
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
vehicle2	93.77 ±	95.08 ±	97.06 ±	94.64 ±	85.39 ±	84.65 ±	98.88 ±	96.59 ±	96.23 ±	97.91 ±	98.68 ±	98.51 ±	81.26 ±	82.98 ±	96.17 ±
	3.88 ↓	4.92 ↓	4.08 ↓	2.92 ↓	5.39 ↓	5.98 ↓	1.97	2.33 ↓	2.97 ↓	1.75	1.25	1.77	7.69 ↓	8.84 ↓	6.22 ↓
vehicle3	76.29 ±	76.64 ±	98.47 ±	76.95 ±	75.06 ±	79.70 ±	90.06 ±	79.91 ±	92.79 ±	75.77 ±	80.01 ±	92.01 ±	74.11 ±	78.61 ±	91.77 ±
	3.78 ↓	3.98 ↓	2.33 ↑	4.20	3.57 ↓	2.96	8.65	3.05	6.22	3.69 -	6.19	9.61	3.57 ↓	5.62	11.75
wisconsin	93.11 ±	92.81 ±	97.40 ±	95.95 ±	95.51 ±	96.32 ±	96.84 ±	98.05 ±	95.72 ±	97.0 ↓	98.08 ±	97.37 ±	93.27 ±	94.46 ±	95.48 ±
	4.42 ↓	5.85 ↓	2.20	2.12	2.39 ↓	2.17 ↓	2.76	1.91	3.05 ↓	2.01 ↑	1.81 ↑	2.65	4.00 -	4.55 -	3.99
yeast1	71.05 ±	71.18 ±	99.55 ±	75.65 ±	74.55 ±	76.53 ±	93.42 ±	79.33 ±	89.23 ±	74.53 ±	79.00 ±	89.31 ±	73.56 ±	83.15 ±	79.18 ±
	3.50 ↓	3.44 ↓	0.80 ↑	2.67 v	2.88 ↓	4.03 ↓	5.28	2.90	4.66 ↓	3.53 ↓	6.12	8.37	4.87 ↓	4.33 ↑	7.29 ↓
yeast3	89.23 ±	89.26 ±	99.95 ±	94.90 ±	89.24 ±	89.25 ±	99.97 ±	97.07 ±	97.22 ±	95.53 ±	98.20 ±	96.77 ±	94.00 ±	98.29 ±	94.98 ±
	2.02 ↓	2.07 ↓	0.19 ↑	1.66	1.30 ↓	1.37 ↓	0.14 ↑	1.23 ↑	1.59 ↓	1.53 ↑	1.20 ↑	1.40 ↓	2.28 ↓	2.14 ↑	2.36 ↓
Average	84.48 ±	83.12 ±	90.54 ±	89.16 ±	88.07 ±	88.51 ±	92.26 ±	90.41 ±	90.92 ±	90.47 ±	92.27 ±	92.75 ±	86.58 ±	88.40 ±	89.19 ±
	4.72	6.64	5.22	3.71	3.56	4.46	4.34	4.28	5.03	3.46	3.62	4.60	6.10	6.83	7.90

Table A2-1

Accuracy, precision and recall values of the used algorithms for BAG, XGB, LGBM and CATB (continued).

Dataset	XGB-RC+			XGB			CATB-RC+			CATB		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
australian	84.49 ± 3.02 ↓	81.67 ± 6.10 ↓	85.37 ± 8.58	86.81 ± 3.81	85.33 ± 4.93	85.39 ± 6.34	85.46 ± 2.96 ↓	83.88 ± 5.69	84.29 ± 7.55	86.67 ± 3.50	84.90 ± 5.19	85.69 ± 6.08
bands	64.03 ± 2.89 ↓	64.95 ± 4.47 ↓	95.94 ± 8.91 ↑	72.08 ± 5.16	74.64 ± 4.01	84.64 ± 7.29	63.47 ± 3.71 ↓	64.75 ± 4.07 ↓	94.49 ± 10.77 ↑	73.53 ± 5.07	73.65 ± 3.76	90.58 ± 7.05
banknote	98.66 ± 1.46 ↓	98.97 ± 1.09 ↓	98.03 ± 3.23 ↓	99.51 ± 0.72	99.26 ± 1.34	99.67 ± 0.79	98.69 ± 1.32 ↓	98.38 ± 1.57 ↓	98.69 ± 2.13 ↓	99.78 ± 0.34	99.52 ± 0.75	100.00 ± 0.00
biodeg	83.57 ± 5.10 ↓	88.71 ± 5.13	86.75 ± 5.27 ↓	86.67 ± 3.46	88.80 ± 2.98	91.51 ± 3.80	83.89 ± 4.01 ↓	88.50 ± 4.52	87.37 ± 4.57 ↓	86.95 ± 3.52	88.85 ± 3.23	91.94 ± 3.56
cardiotcg	94.83 ± 1.29 ↓	95.79 ± 1.27 ↓	97.66 ± 1.38 ↓	95.83 ± 1.23	96.51 ± 1.09	98.21 ± 1.20	94.26 ± 1.55 ↓	95.99 ± 1.18	96.68 ± 1.69 ↓	95.63 ± 1.37	96.28 ± 1.20	98.19 ± 1.23
credit	99.70 ± 0.61	96.83 ± 6.48	100.00 ± 0.00	99.70 ± 0.61	96.83 ± 6.48	100.00 ± 0.00	99.85 ± 0.46	98.41 ± 4.86	100.00 ± 0.00	99.85 ± 0.46	98.41 ± 4.86	100.00 ± 0.00
cryotherapy	89.63 ± 8.22	95.92 ± 9.52	85.83 ± 13.78	91.11 ± 9.41	95.10 ± 9.24	88.83 ± 14.06	88.89 ± 9.23	93.60 ± 11.17	86.83 ± 14.71	91.85 ± 8.72	97.94 ± 6.37	86.67 ± 13.85
divorce	96.67 ± 3.68	98.18 ± 4.17	95.14 ± 6.07	97.65 ± 2.93	100.00 ± 0.00	95.19 ± 6.01	96.67 ± 3.99	97.93 ± 4.98	95.65 ± 5.83	97.65 ± 2.93	100.00 ± 0.00	95.19 ± 6.01
ecoli0vs1	98.64 ± 2.12	98.04 ± 3.04	100.00 ± 0.00	98.64 ± 2.12	98.04 ± 3.04	100.00 ± 0.00	98.48 ± 2.48	98.26 ± 2.93	99.52 ± 2.61	98.64 ± 2.43	98.08 ± 3.36	100.00 ± 0.00
ecoli1	88.68 ± 5.09	92.98 ± 5.41	92.79 ± 5.72	89.68 ± 4.66	92.93 ± 4.61	94.09 ± 5.22	87.03 ± 5.08 ↓	93.23 ± 4.64	90.10 ± 7.47 ↓	90.39 ± 4.64	92.96 ± 4.66	94.99 ± 4.06
ecoli2	94.05 ± 2.57	95.62 ± 2.86	97.53 ± 2.48	94.65 ± 3.05	95.73 ± 2.91	98.13 ± 2.19	93.38 ± 4.68 ↓	95.06 ± 4.26 ↓	97.43 ± 3.43	95.05 ± 2.72	96.15 ± 2.58	98.14 ± 1.99
ecoli3	91.26 ± 4.24	94.56 ± 3.24	95.91 ± 4.25 ↓	92.56 ± 3.49	94.61 ± 2.83	97.35 ± 3.07	91.17 ± 4.23	93.96 ± 3.69	96.58 ± 4.40	92.67 ± 2.90	93.97 ± 2.91	98.24 ± 2.42
glass	91.60 ± 5.20 ↓	95.16 ± 4.92	94.07 ± 5.48	93.30 ± 5.00	95.67 ± 3.96	95.71 ± 5.40	91.59 ± 6.10	94.82 ± 5.82	94.66 ± 6.66	93.92 ± 4.46	96.06 ± 3.48	96.10 ± 5.26
glass0	83.07 ± 11.13	89.35 ± 7.88	85.48 ± 13.47 ↓	85.21 ± 8.55	88.69 ± 7.13	90.03 ± 9.45	84.22 ± 9.06 ↓	88.80 ± 9.28	89.03 ± 8.94	87.83 ± 7.84	91.39 ± 6.27	90.70 ± 8.37
glass1	79.24 ± 10.78	83.95 ± 10.40	86.48 ± 12.45	82.52 ± 7.16	86.04 ± 6.84	87.93 ± 8.50	80.61 ± 10.08	83.68 ± 9.00	87.82 ± 10.85 ↓	83.76 ± 8.06	84.19 ± 8.21	93.44 ± 6.47
glass6	94.39 ± 4.20 ↓	96.44 ± 3.81	97.29 ± 3.99	97.06 ± 4.16	97.77 ± 3.22	98.94 ± 2.16	94.57 ± 5.47 ↓	96.55 ± 3.63	97.31 ± 4.64	96.90 ± 3.54	97.62 ± 3.19	98.93 ± 2.18
haberman	72.97 ± 3.54	73.70 ± 1.17 ↓	98.52 ± 5.64 ↑	73.83 ± 7.37	78.07 ± 4.65	89.95 ± 6.54	72.86 ± 4.29	74.16 ± 2.42 ↓	97.19 ± 6.42 ↑	74.61 ± 5.88	77.63 ± 3.73	92.35 ± 5.19
heart	78.02 ± 9.07	81.97 ± 9.91	79.56 ± 11.74 ↓	81.36 ± 7.24	82.56 ± 7.66	85.11 ± 8.15	79.26 ± 9.00 ↓	81.31 ± 9.63 ↓	83.56 ± 11.04 -	85.06 ± 6.84	85.31 ± 7.11	88.89 ± 8.09
ionosphere	89.17 ± 5.53 ↓	89.84 ± 6.76 ↓	94.49 ± 5.11 ↓	92.60 ± 3.91	92.40 ± 4.40	96.59 ± 2.78	90.13 ± 5.32 ↓	91.41 ± 5.09 ↓	93.61 ± 5.09 ↓	93.54 ± 4.17	92.71 ± 4.99	97.92 ± 2.82
maintenance	98.43 ± 0.52	86.60 ± 10.46	64.52 ± 7.85 ↓	98.59 ± 0.34	87.57 ± 7.71	99.87 ± 7.14	98.04 ± 0.46 ↓	74.20 ± 11.32 ↓	68.34 ± 9.44	98.58 ± 0.31	89.46 ± 5.39	65.79 ± 7.19
mammographic	83.05 ± 4.87	83.37 ± 6.59 ↓	81.95 ± 7.50	84.10 ± 4.24	85.67 ± 4.94	80.88 ± 5.83	82.93 ± 4.11 ↓	84.38 ± 5.71	80.13 ± 7.98	84.30 ± 4.24	84.99 ± 4.93	82.29 ± 5.84
newthyroid1	97.53 ± 2.89	98.41 ± 2.47	98.70 ± 2.80	97.37 ± 2.91	97.55 ± 2.97	99.44 ± 1.70	96.59 ± 3.22 ↓	97.90 ± 2.93	98.15 ± 3.37 ↓	98.90 ± 2.02	98.77 ± 2.26	100.00 ± 0.00
newthyroid2	96.93 ± 3.02	98.43 ± 2.77	97.96 ± 3.09	97.53 ± 3.16	97.77 ± 3.45	99.44 ± 1.70	96.59 ± 4.26 ↓	98.14 ± 3.55	97.96 ± 4.25 ↓	98.61 ± 2.98	98.47 ± 3.22	100.00 ± 0.00
occupancy	99.28 ± 0.29	99.61 ± 0.32	99.48 ± 0.35	99.37 ± 0.24	99.66 ± 0.21	99.54 ± 0.25	99.09 ± 0.39 ↓	99.68 ± 0.29	99.16 ± 0.59 ↓	99.42 ± 0.22	99.72 ± 0.18	99.54 ± 0.26
pageblocks	97.31 ± 0.72 ↓	98.61 ± 0.67	98.40 ± 0.83 ↓	97.59 ± 0.59	98.52 ± 0.44	98.80 ± 0.52	97.28 ± 0.50 ↓	98.49 ± 0.48	98.48 ± 0.69 ↓	97.58 ± 0.61	98.51 ± 0.43	98.81 ± 0.54
pima	74.08 ± 5.40 ↓	79.03 ± 4.56	82.67 ± 8.60	75.77 ± 4.46	79.34 ± 3.64	85.13 ± 5.32	73.92 ± 4.85	78.96 ± 6.06	83.27 ± 8.65	75.56 ± 4.17	78.58 ± 3.66	86.13 ± 4.30
segment0	99.42 ± 0.53 ↓	99.53 ± 0.47 ↓	99.80 ± 0.34	99.70 ± 0.33	99.78 ± 0.31	99.87 ± 0.26	99.54 ± 0.41 ↓	99.71 ± 0.34 ↓	99.75 ± 0.35 ↓	99.78 ± 0.32	99.80 ± 0.28	99.95 ± 0.15
tic-tac-toe	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
vehicle2	97.75 ± 1.93 ↓	98.69 ± 1.37	98.30 ± 2.09 ↓	98.38 ± 1.50	98.75 ± 1.25	99.10 ± 1.43	97.04 ± 2.19 ↓	98.10 ± 1.48 ↓	97.93 ± 2.44 ↓	98.34 ± 1.47	99.00 ± 0.97	98.78 ± 1.54
vehicle3	75.77 ± 2.84 ↓	77.48 ± 3.79 ↓	96.11 ± 5.55 ↑	79.12 ± 4.65	83.48 ± 2.88	90.02 ± 4.99	76.12 ± 2.99 ↓	79.14 ± 4.63 ↓	93.63 ± 7.56	78.33 ± 3.27	80.46 ± 2.59	94.06 ± 3.72
wisconsin	96.00 ± 2.09	97.83 ± 2.04	96.02 ± 2.76	96.29 ± 2.09	97.57 ± 2.36	96.77 ± 2.56	95.51 ± 3.20 ↓	96.99 ± 2.91	96.16 ± 3.90	96.83 ± 1.99	97.80 ± 2.20	97.37 ± 2.52
yeast1	74.41 ± 3.50 ↓	77.69 ± 5.39 ↓	91.35 ± 7.51	78.37 ± 3.25	81.13 ± 2.56	90.78 ± 2.87	75.09 ± 3.71 ↓	78.62 ± 4.04 ↓	89.95 ± 5.81 ↓	78.19 ± 2.87	80.51 ± 2.13	91.53 ± 2.77
yeast3	95.01 ± 1.81	97.71 ± 1.38	96.69 ± 1.94	95.01 ± 1.42	97.25 ± 1.31	97.17 ± 1.35	95.13 ± 1.72	97.56 ± 1.48 ↑	96.97 ± 1.53 ↓	95.13 ± 1.57	96.96 ± 1.25	97.60 ± 1.30
Average	89.63 ± 3.64	91.08 ± 4.24	92.99 ± 5.11	91.15 ± 3.43	92.21 ± 3.50	93.42 ± 3.91	89.62 ± 3.79	90.74 ± 4.35	93.05 ± 5.31	91.63 ± 3.19	92.38 ± 3.19	94.24 ± 3.48

Table A2-2

Accuracy, precision and recall values of the used algorithms for BAG, XGB, LGBM and CATB.

Dataset	LGBM-RC+ Accuracy	Precision	Recall	LGBM Accuracy	Precision	Recall
australian	85.51 ± 2.72	83.77 ± 4.12	84.19 ± 7.10	86.14 ± 3.11	83.42 ± 4.26	86.23 ± 4.52
bands	63.21 ± 1.12 ↓	63.78 ± 2.63 ↓	97.68 ± 7.29 ↑	69.32 ± 6.88	72.04 ± 5.36	84.49 ± 8.52
banknote	98.54 ± 1.65 ↓	98.28 ± 2.01 ↓	98.47 ± 2.69 ↓	99.73 ± 0.40	99.52 ± 0.75	99.89 ± 0.42
biodeg	84.37 ± 3.94 ↓	88.46 ± 3.04	87.94 ± 4.36 ↓	87.39 ± 3.30	89.50 ± 3.06	91.85 ± 3.33
cardiotcg	94.79 ± 1.54 ↓	95.97 ± 1.28 ↓	97.42 ± 1.72 ↓	95.97 ± 1.06	96.61 ± 1.13	98.29 ± 1.19
credit	99.70 ± 0.61	96.83 ± 6.48	100.00 ± 0.00	99.70 ± 0.61	96.83 ± 6.48	100.00 ± 0.00
cryotherapy	84.81 ± 10.31 ↓	96.16 ± 8.91	75.33 ± 19.78 ↓	89.26 ± 9.45	92.33 ± 10.56	88.00 ± 14.12
divorce	96.67 ± 3.99	98.02 ± 5.37	95.51 ± 6.68	97.45 ± 2.96	99.63 ± 2.03	95.19 ± 6.01
ecoli0vs1	98.18 ± 2.56	98.28 ± 3.34	99.06 ± 2.43	98.33 ± 2.53	98.28 ± 3.34	99.30 ± 2.13
ecoli1	88.19 ± 5.99	91.65 ± 5.06	93.54 ± 7.63	89.10 ± 4.92	91.56 ± 4.99	94.99 ± 5.46
ecoli2	94.25 ± 3.08	96.22 ± 2.54	97.06 ± 2.93 ↓	95.15 ± 2.95	96.08 ± 2.89	98.37 ± 2.53
ecoli3	91.36 ± 3.69 ↓	93.33 ± 3.05	97.46 ± 3.46	93.16 ± 2.71	94.45 ± 2.63	98.24 ± 2.56
glass	91.88 ± 5.06 ↓	96.40 ± 3.96	93.08 ± 6.55 ↓	94.40 ± 4.29	96.30 ± 3.83	96.52 ± 4.48
glass0	80.80 ± 11.18 ↓	86.97 ± 8.71 ↓	84.71 ± 13.10 ↓	84.76 ± 8.26	89.41 ± 6.55	88.22 ± 9.50
glass1	80.82 ± 9.57	84.89 ± 9.77	87.27 ± 9.56	82.19 ± 8.27	85.29 ± 8.45	88.61 ± 7.69
glass6	94.40 ± 4.98 ↓	95.69 ± 4.69	98.22 ± 2.57	96.91 ± 3.49	97.46 ± 3.16	99.11 ± 2.02
haberman	72.98 ± 3.43	73.99 ± 1.45 ↓	97.91 ± 7.04 ↑	71.98 ± 5.16	75.46 ± 3.36	92.31 ± 7.59
heart	78.64 ± 9.61	81.55 ± 10.38	81.78 ± 10.49 ↓	81.60 ± 6.56	81.83 ± 7.72	87.11 ± 6.77
ionosphere	91.16 ± 4.59 -	91.51 ± 5.07	95.39 ± 4.51 ↓	93.07 ± 4.29	92.44 ± 5.22	97.47 ± 2.81
maintenance	98.2 ↓ ± 0.46 -	80.21 ± 10.58 -	65.61 ± 6.71	98.61 ± 0.30	89.67 ± 5.61	66.79 ± 7.50
mammographic	83.6 ↓ ± 4.11	84.77 ↓ ± 5.43	81.37 ± 6.63	84.26 ± 4.29	85.83 ± 4.95	81.05 ± 5.81
newthyroid1	96.92 ± 3.69	98.08 ± 3.20	98.33 ± 2.97	97.51 ± 2.68	98.24 ± 2.54	98.89 ± 2.26
newthyroid2	97.39 ± 3.35	98.59 ± 2.71	98.33 ± 2.97	97.08 ± 3.07	97.93 ± 3.18	98.70 ± 2.80
occupancy	99.12 ± 0.33 ↓	99.82 ± 0.20 ↑	99.06 ± 0.43 ↓	99.34 ± 0.23	99.66 ± 0.21	99.50 ± 0.24
pageblocks	97.12 ± 0.66 ↓	98.66 ± 0.67	98.13 ± 0.89 ↓	97.53 ± 0.60	98.45 ± 0.45	98.81 ± 0.48
pima	73.99 ± 4.84	78.74 ± 5.08 ↑	83.20 ± 7.71 ↓	75.12 ± 3.72	76.70 ± 3.21	89.13 ± 5.75
segment0	99.48 ± 0.43 ↓	99.61 ± 0.34 ↓	99.78 ± 0.37	99.77 ± 0.32	99.85 ± 0.23	99.88 ± 0.25
tic-tac-toe	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
vehicle2	97.60 ± 1.87 ↓	98.22 ± 1.44 ↓	98.57 ± 2.10	98.70 ± 1.36	99.26 ± 0.99	98.99 ± 1.53
vehicle3	75.14 ± 2.54 ↓	77.89 ± 3.55 ↓	94.14 ± 7.98	79.28 ± 4.43	81.95 ± 4.15	93.27 ± 5.55
wisconsin	95.85 ± 2.21	97.59 ± 2.03	96.02 ± 2.81	96.29 ± 2.09	97.48 ± 1.96	96.84 ± 2.62
yeast1	74.42 ± 3.82 ↓	77.66 ± 5.18	91.26 ± 7.30	77.36 ± 3.45	79.85 ± 2.85	91.31 ± 3.06
yeast3	95.06 ± 1.82	97.51 ± 1.40	96.95 ± 1.81 ↓	95.19 ± 1.69	97.01 ± 1.40	97.63 ± 1.31
Average	89.52 ± 3.63	90.88 ± 4.05	92.81 ± 5.17	90.96 ± 3.32	91.83 ± 3.56	93.79 ± 3.96

Appendix B. Distribution of hyperparameter values

Appendix para(Figure B1, Figure B2, Figure B3, Figure B4, Figure B5, Figure B6)

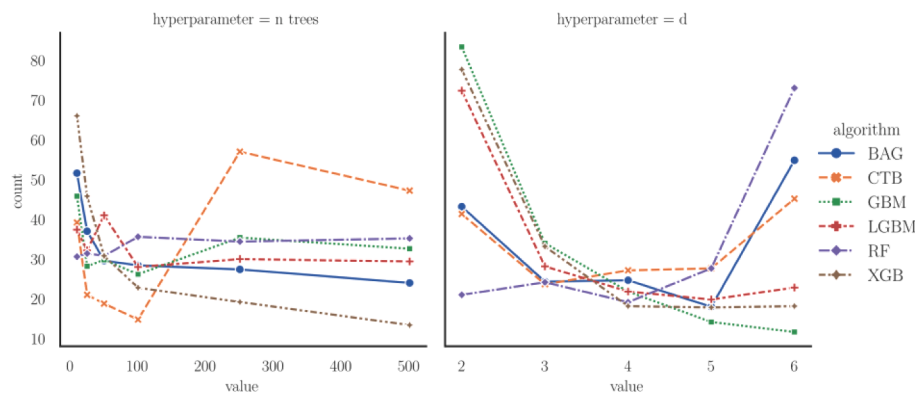


Fig. B1. Distribution of the best hyperparameter values of original ensembles, (a) n_trees and (b) depth.

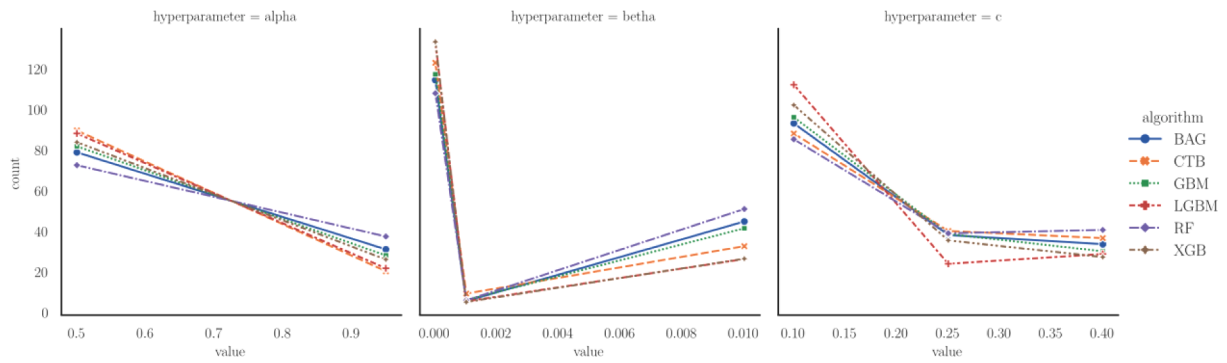


Fig. B2. Distribution of the best hyperparameter values of RuleCOSI+, (a) alpha, (b) beta, and (c) confidence level.

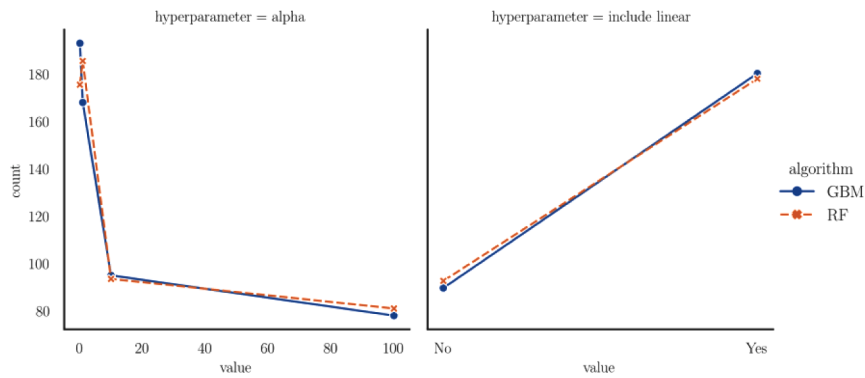


Fig. B3. Distribution of the best hyperparameter values of RuleFit, (a) alpha and (b) linear.

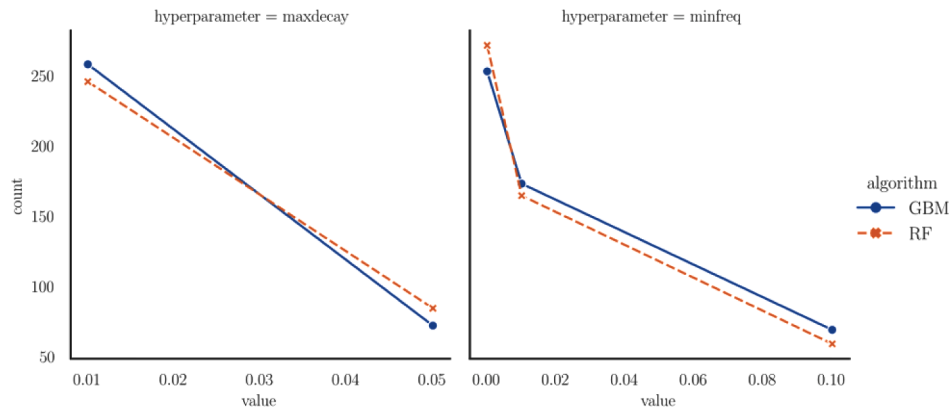


Fig. B4. Distribution of the best hyperparameter values of inTrees, (a) maxdecay and (b) minfreq.

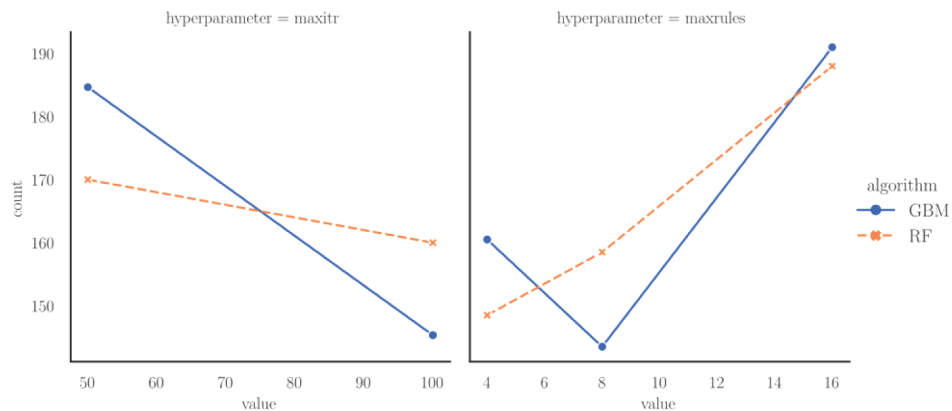


Fig. B5. Distribution of the best hyperparameter values of DefragTrees, (a) maxitr and (b) maxrules.

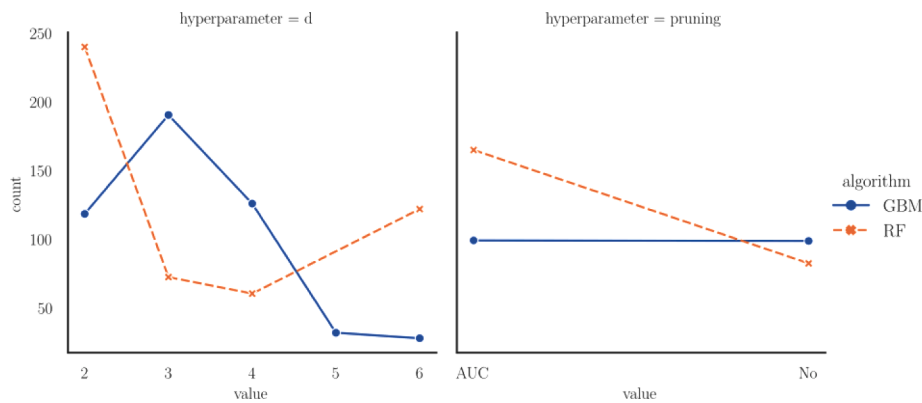


Fig. B6. Distribution of the best hyperparameter values of FBT, (a) depth and (b) pruning.

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